

# A CRM Application For Public Transport Management

## A CRM APPLICATION FOR PUBLIC TRANSPORT MANAGEMENT SYSTEM

**COLLEGE NAME:- GAYATRI DEGREE COLLEGE TIRUPATI**

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# A CRM Application For Public Transport Management

## **Introduction: -**

The Public Transport (RTC - Regional Transport corporation) Management System is a comprehensive Salesforce application designed to streamline and manage various operational aspects of the Public Transport. This system will enable the Transport department to efficiently maintain details of employees, their roles, bus stations, buses, ticket fares, daily bus trips, passenger counts, and the total ticket fare amount. By leveraging Salesforce's robust platform, the Transport Department can improve operational efficiency, data accuracy, and reporting capabilities.

## **Project Overview:**

This project is to Consolidate all Transport-related data into a single Salesforce application to ensure easy access, management, and reporting. Enable real-time insights and reports on various operational metrics such as passenger count and revenue. Maintain detailed records of all Transport employees, including personal details, contact information. Define and assign roles and responsibilities to each employee. Schedule and manage employee shifts, especially for drivers and conductors. Maintain information on all bus stations, including location, facilities. Maintain detailed records of all buses, including model, capacity. Manage bus schedules and assign buses to specific routes and trips. Define and manage ticket fares for different routes and bus types. Track daily trips for each bus, including start and end times, routes, and driver details. Record the number of passengers on each trip. Calculate and track the total ticket fare amount collected from each trip. Provide real-time dashboards for quick insights into key metrics such as passenger count, trip efficiency, and revenue.

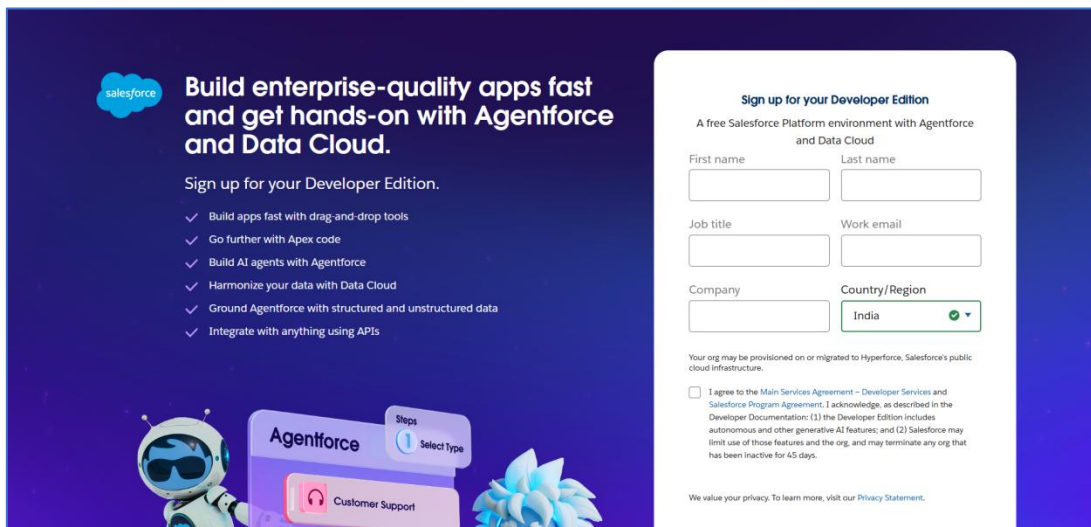
# A CRM Application For Public Transport Management

## Milestone 1 Salesforce

### Activity 1: Creating Developer Account

Creating a developer org in Salesforce

1. Go to <https://developer.salesforce.com/signup>
2. On the sign up form, enter the following details :



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Job title  Work email

Company  Country/Region

Your org may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

☐ I agree to the Main Services Agreement – Developer Services and Salesforce Program Agreement. I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

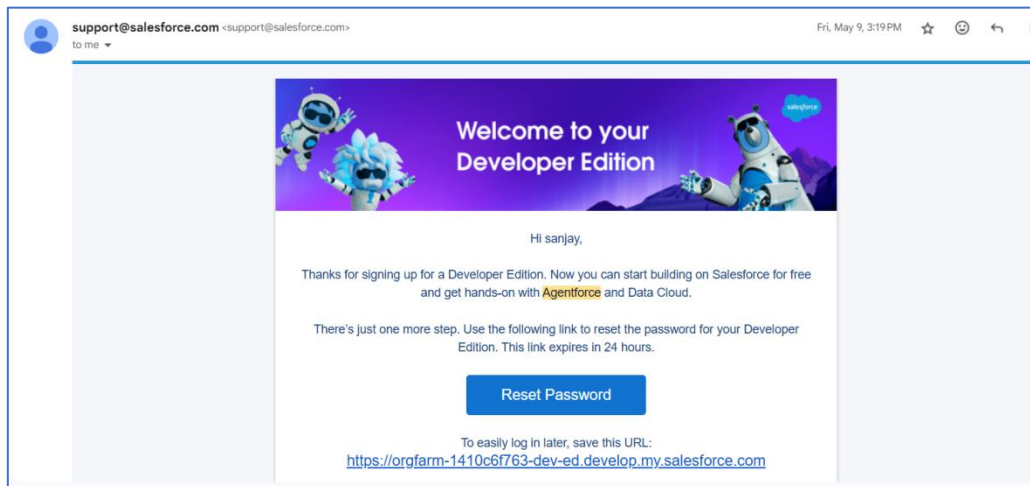
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1. First name – POLISETTY
2. Last name - SANJAY KUMAR
3. Email – sanjaypolisettyjanju@gmail.com
4. Role : Developer
5. Company : Gayatri Degree College Tirupati
6. County : India

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## Activity 2: Account Activation

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.

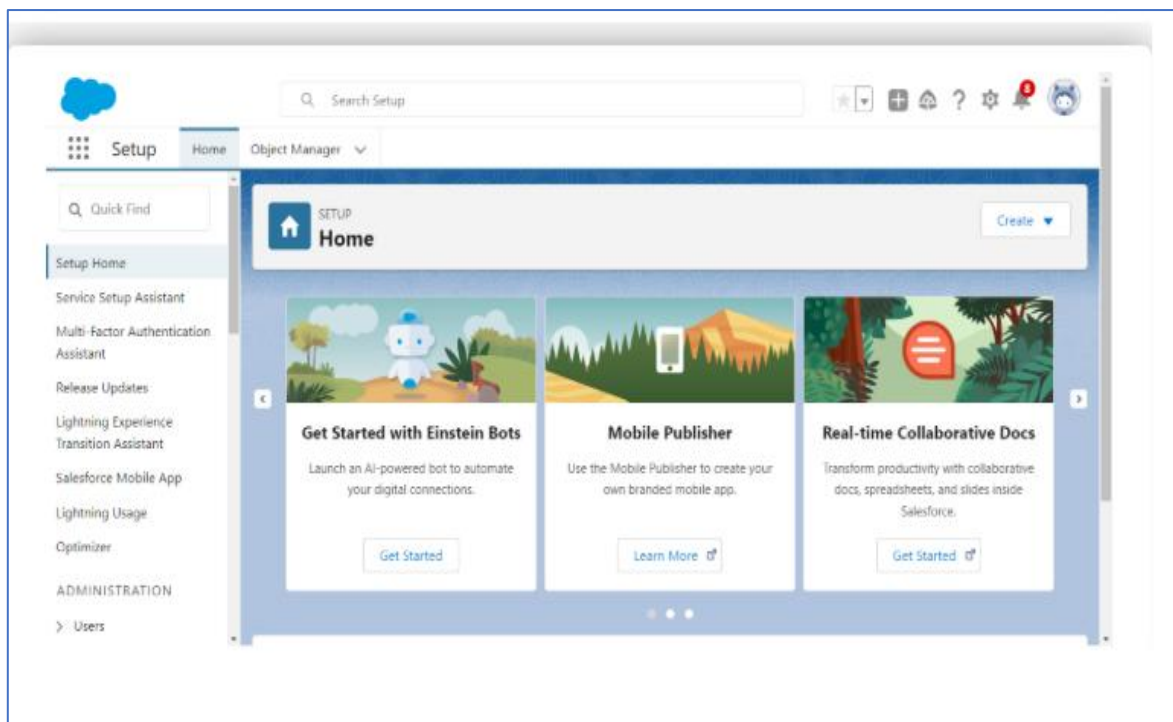


2. Click on Verify Account
3. Give a password and answer a security question and click on change password.

A screenshot of the 'Change Your Password' form in Salesforce. The form is titled 'Change Your Password' and includes instructions to enter a new password for lead@sb.com. It lists requirements for the password: 8 characters, 1 letter, and 1 number. The form has four main sections: 'New Password' with a 'Good' status indicator, 'Confirm New Password' with a 'Match' status indicator, 'Security Question' with a dropdown menu showing 'In what city were you born?', and 'Answer' with a text input field containing 'asdfghjkl'. A red box highlights the 'New Password', 'Confirm New Password', 'Security Question', 'Answer', and 'Change Password' button sections. The 'Change Password' button is a blue button at the bottom of the form.

4. Then you will redirect to your Salesforce setup page.

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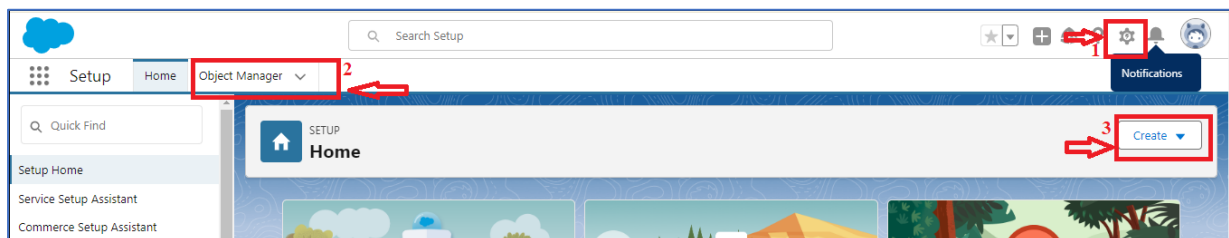
## Milestone 2 Objects

### Activity 1: Creating a Bus Station Object

The purpose of creating a Bus Station custom object is to store and manage information about Bus Stops.

To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.



4. Enter the label name as Bus Station
5. Enter Plural label name as Bus Stations
6. Enter Record Name as Bus Station Name

This screenshot shows the 'New Custom Object' form in the Salesforce Setup interface. The form is titled 'New Custom Object' and has a 'Custom Object Definition Edit' header. The 'Custom Object Information' section contains the following fields:

- Label:** 'Bus Station' (highlighted with a red box and labeled '4'). Example: Account.
- Plural Label:** 'Bus stations' (highlighted with a red box and labeled '5'). Example: Accounts.
- Object Name:** 'Bus\_Station' (highlighted with a red box). Example: Account.
- Description:** A text area for the object's description.
- Context-Sensitive Help Setting:** A radio button selection between 'Open the standard Salesforce.com Help & Training window' (selected) and 'Open a window using a Visualforce page'.
- Content Name:** A dropdown menu with 'None' selected.

The form also includes 'Save', 'Save & New', and 'Cancel' buttons at the top right of the form area.

# A CRM Application For Public Transport Management

7. Select Data Type as Text.
8. Select Allow reports.

Setup Home Object Manager

## SETUP New Custom Object

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

6 Record Name Bus Station Name example: Account Name

7 Data Type Text

### Optional Features

8 ☒ Allow Reports

☐ Allow Activities

☐ Track Field History

☐ Allow in Chatter Groups

☐ Enable Licensing

### Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

☒ Allow Sharing

☒ Allow Bulk API Access

☒ Allow Streaming API Access

### Deployment Status

☐ In Development

☒ Deployed

Activate Windows  
Go to Settings to activate Windows.

[What is this?](#)

9. Select Allow search.
10. Click on Save and New

### Deployment Status

☐ In Development

☒ Deployed

### Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

9 ☒ Allow Search

### Object Creation Options (Available only when custom object is first created)

☐ Add Notes and Attachments related list to default page layout

☐ Launch New Custom Tab Wizard after saving this custom object

10 Save Save & New Cancel

Activate Windows  
Go to Settings to activate Windows.

[What is this?](#)

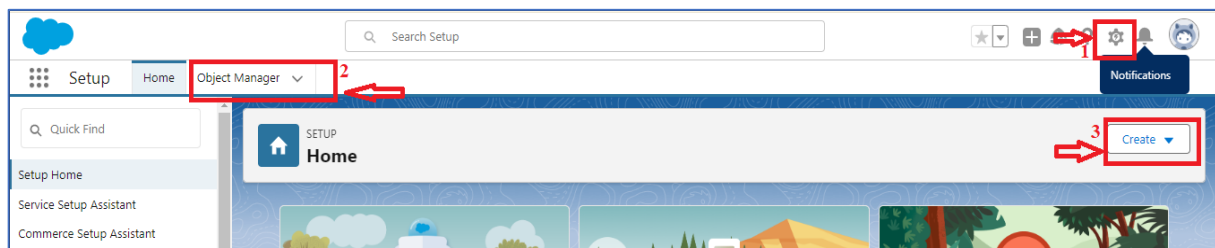
In the same way Create Bus, Trip, Ticket Fare and Employee objects.

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## Activity 2: Creating a Bus Object

The purpose of creating a Bus custom object is to store and manage information about Bus Stops.  
To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.



4. Enter the label name as Bus
5. Enter Plural label name as Buses
6. Enter Record Name as Bus Registration Number

**Custom Object Information**

The singular and plural labels are used in tabs, page layouts, and reports.  
**Be careful when changing the name or label as it may affect existing integrations and merge templates.**

Label  Example: Account

Plural Label  Example: Accounts

Starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name  Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window  
☐ Open a window using a Visualforce page

Content Name

7. Select Data Type as Text.
8. Select Allow reports.



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The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name is called "Name" when referenced via the API.

Record Name:  Example: Account Name

Data Type:  Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

**Optional Features**

- ☒ Allow Reports
- ☐ Allow Activities
- ☐ Track Field History
- ☐ Allow in Chatter Groups
- ☐ Enable Licensing [i](#)

**Object Classification**

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

**Deployment Status**

☐ In Development

☒ Deployed

**Search Status**

9. Select Allow search.
10. Click on Save and New

**Deployment Status**

☐ In Development

☒ Deployed

**Search Status**

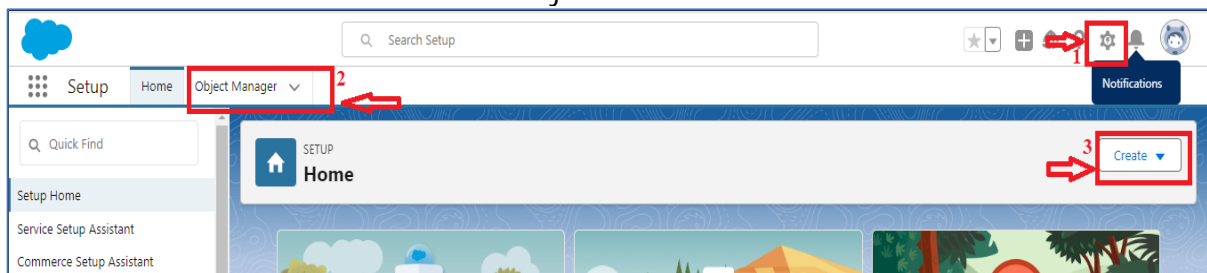
When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

☒ Allow Search

## Activity 3: Creating a Trip Object

The purpose of creating a Bus custom object is to store and manage information about Bus Stops. To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.



# A CRM Application For Public Transport Management

4. Enter the label name as Trip
5. Enter Plural label name as Trips
6. Enter Record Name as Trip No

**Custom Object Definition Edit** [Save] [Save & New] [Cancel]

**Custom Object Information**

The singular and plural labels are used in tabs, page layouts, and reports.  
**Be careful when changing the name or label as it may affect existing integrations and merge templates.**

Label  **Example: Account**

Plural Label  **Example: Accounts**

Starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name  **Example: Account**

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window  
☐ Open a window using a Visualforce page

Content Name

7. Select Data Type as Text.
8. Select Allow reports.

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name  **Example: Account Name**

Data Type  **Warning:** If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

**Optional Features**

☒ Allow Reports  
☐ Allow Activities  
☐ Track Field History  
☐ Allow in Chatter Groups  
☐ Enable Licensing [i](#)

**Object Classification**

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

☒ Allow Sharing  
☒ Allow Bulk API Access  
☒ Allow Streaming API Access

9. Select Allow search.
10. Click on Save and New

**Deployment Status**

☐ In Development  
☒ Deployed

**Search Status**

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

☒ Allow Search

[Save] [Save & New] [Cancel]

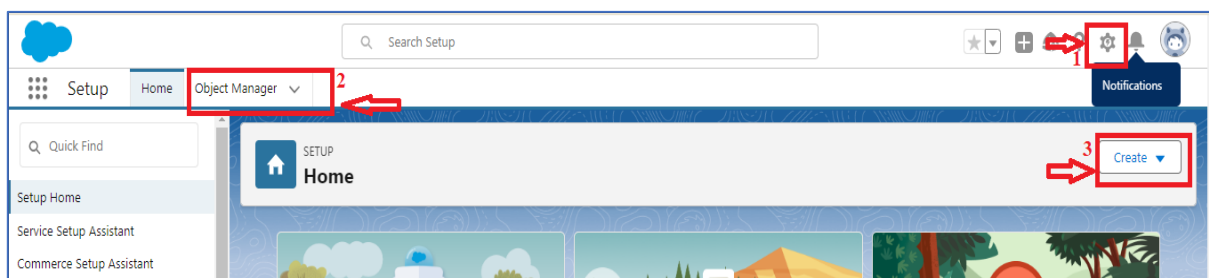
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## Activity 4: Creating a Ticket Fare Object

The purpose of creating a Ticket Fare custom object is to store and manage information about Bus Stops.

To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.



4. Enter the label name as Ticket Fare
5. Enter Plural label name as Ticket Fares
6. Enter Record Name as Route Name

**Custom Object Definition Edit** [Save] [Save & New] [Cancel]

**Custom Object Information**

The singular and plural labels are used in tabs, page layouts, and reports.  
**Be careful when changing the name or label as it may affect existing integrations and merge templates.**

Label:  Example: Account

Plural Label:  Example: Accounts

Starts with vowel sound: ☐

The Object Name is used when referencing the object via the API.

Object Name:  Example: Account

Description:

Context-Sensitive Help Setting: ☒ Open the standard Salesforce.com Help & Training window  
☐ Open a window using a Visualforce page

Content Name:

7. Select Data Type as Text.
8. Select Allow reports.

# A CRM Application For Public Transport Management

**Enter Record Name Label and Format**

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name  **Example: Account Name**

Data Type  **Warning:** If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

**Optional Features**

- ☒ Allow Reports
- ☐ Allow Activities
- ☐ Track Field History
- ☐ Allow in Chatter Groups
- ☐ Enable Licensing [i](#)

**Object Classification**

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

9. Select Allow search.
10. Click on Save and New

**Deployment Status**

☐ In Development

☒ Deployed

**Search Status**

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

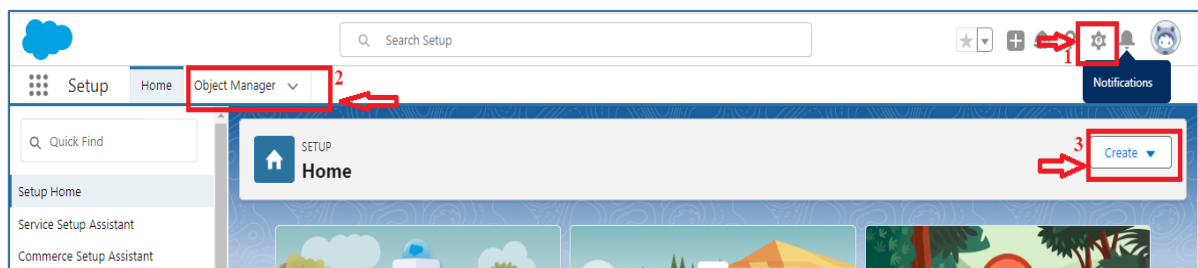
☒ Allow Search

## Activity 5: Creating aEmployee Object

The purpose of creating aEmployee custom object is to store and manage information about Bus Stops.

To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.



4. Enter the label name as Employee
5. Enter Plural label name as Employees
6. Enter Record Name as Employee Id

# A CRM Application For Public Transport Management

**Custom Object Information**

The singular and plural labels are used in tabs, page layouts, and reports. **Be careful when changing the name or label as it may affect existing integrations and merge templates.**

Label  Example: Account

Plural Label  Example: Accounts

Starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name  Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window  
☐ Open a window using a Visualforce page

Content Name

7. Select Data Type as Text.

8. Select Allow reports.

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name is called "Name" when referenced via the API.

Record Name  Example: Account Name

Data Type  **Warning:** If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

**Optional Features**

- ☒ Allow Reports
- ☐ Allow Activities
- ☐ Track Field History
- ☐ Allow in Chatter Groups
- ☐ Enable Licensing [i](#)

**Object Classification**

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

9. Select Allow search.

10. Click on Save and New

**Deployment Status**

☐ In Development

☒ Deployed

**Search Status**

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

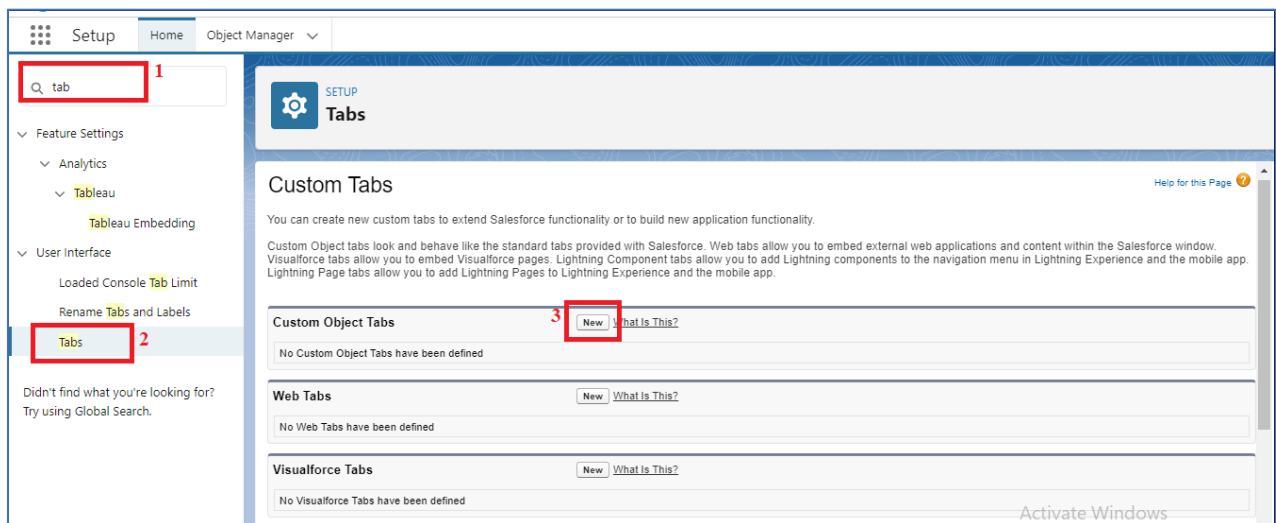
☒ Allow Search

# A CRM Application For Public Transport Management

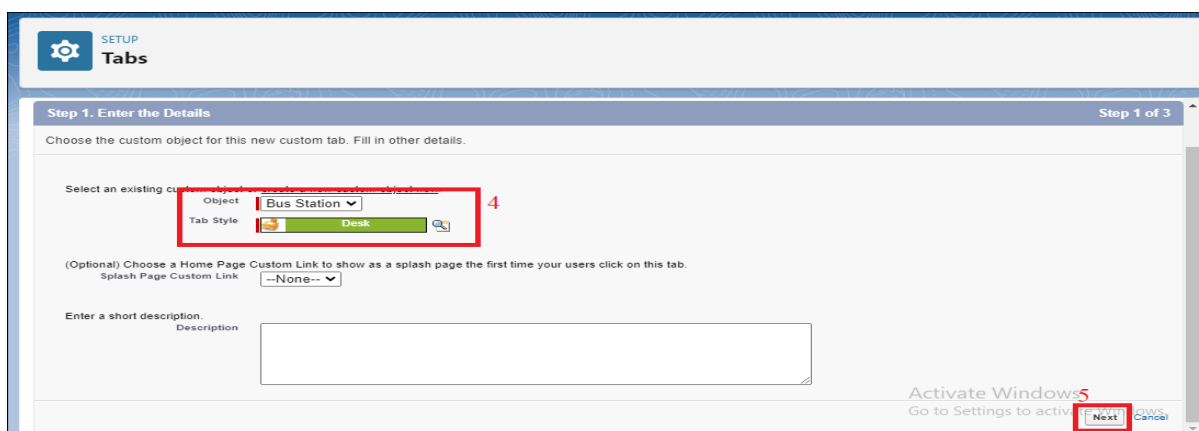
## Milestone 3 Tabs

### Activity 1: Creating a tab for Bus Station Object

1. Go to the setup page >> type Tabs in Quick Find bar
2. Click on tabs
3. Click on New (under custom object tab).



4. Select Object(Bus Station) >> Select the tab style
5. Click on Next >> (Add to profiles page) keep it as default >> Click on Next (Add to Custom App) uncheck the include tab .



6. Make sure that the Append tab to user's existing personal customizations is checked

# A CRM Application For Public Transport Management

7. Click on Save.

**1. Now create the Tabs for the remaining Objects, they are “Bus, Trip, Ticket Fare, Employee”.**

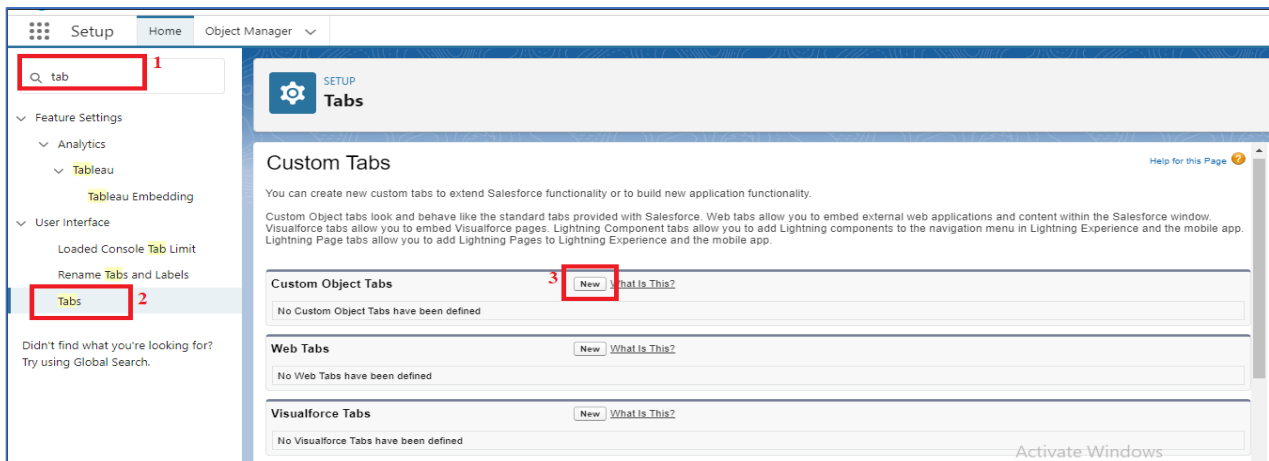
**2. Follow the same steps as mentioned in Activity -1**

## Activity 2: Creating Tabs for Bus Object

1. Go to the setup page >> type Tabs in Quick Find bar

2. Click on tabs

3. Click on New (under custom object tab).



4. Select Object(Bus ) >> Select the tab style

5. Click on Next >>(Add to profiles page) keep it as default >>Click on Next (Add to Custom App) uncheck the include tab .

**Custom Object Tab Information**

|           |                                                                                         |
|-----------|-----------------------------------------------------------------------------------------|
| Tab Label | Buses                                                                                   |
| Object    | Bus                                                                                     |
| Tab Style |  Car |

(Optional) Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.

Splash Page Custom Link --None--

Enter a short description.

Description

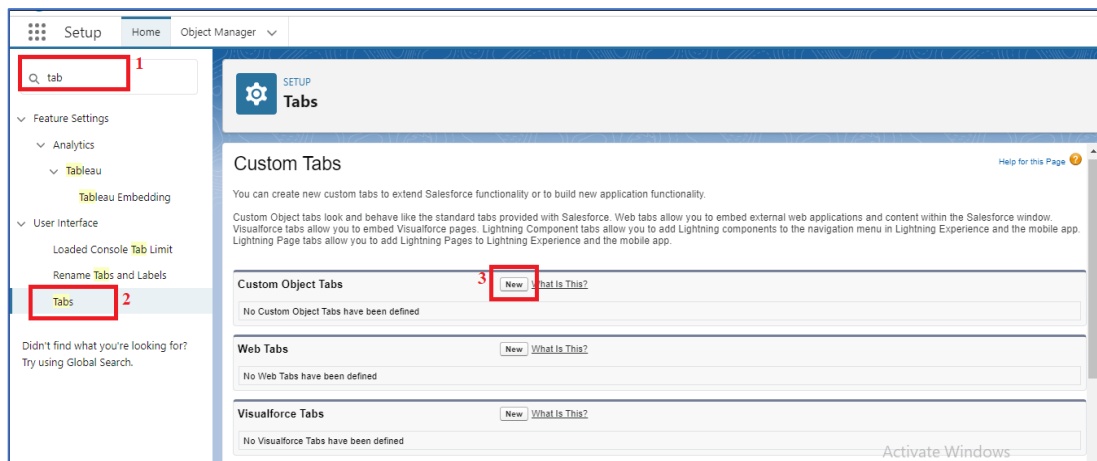
Save Cancel

# A CRM Application For Public Transport Management

6. Make sure that the Append tab to user's existing personal customizations is checked
7. Click on Save.

## Activity 3: Creating Tabs for Trip Object

1. Go to the setup page >> type Tabs in Quick Find bar
2. Click on tabs
3. Click on New (under custom object tab).



4. Select Object(Trip ) >> Select the tab style
5. Click on Next >>(Add to profiles page) keep it as default >>Click on Next (Add to Custom App) uncheck the include tab .

6. Make sure that the Append tab to user's existing personal customizations is checked

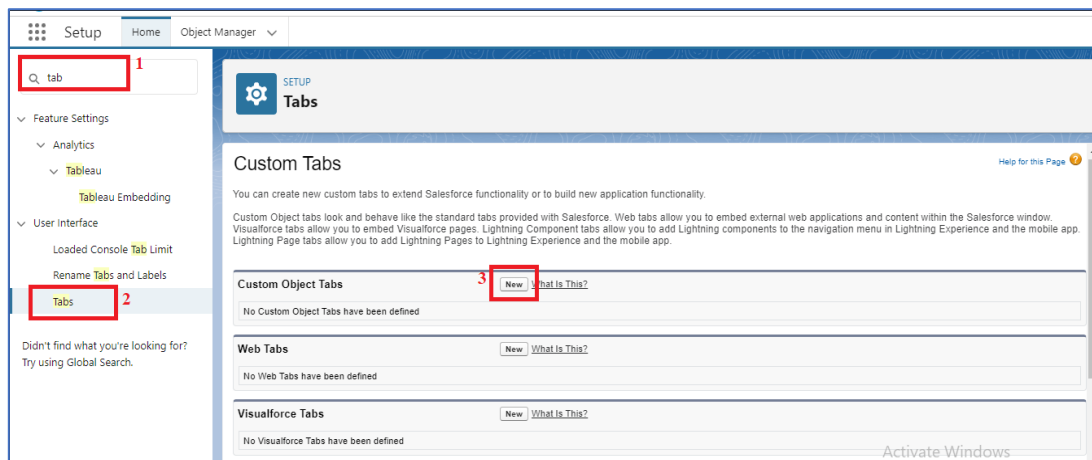


# A CRM Application For Public Transport Management

7. Click on Save.

## Activity 4: Creating Tabs for Ticket Fare Object

1. Go to the setup page >> type Tabs in Quick Find bar
2. Click on tabs
3. Click on New (under custom object tab).



4. Select Object(Ticket Fare ) >> Select the tab style
5. Click on Next >>(Add to profiles page) keep it as default >>Click on Next (Add to Custom App) uncheck the include tab .

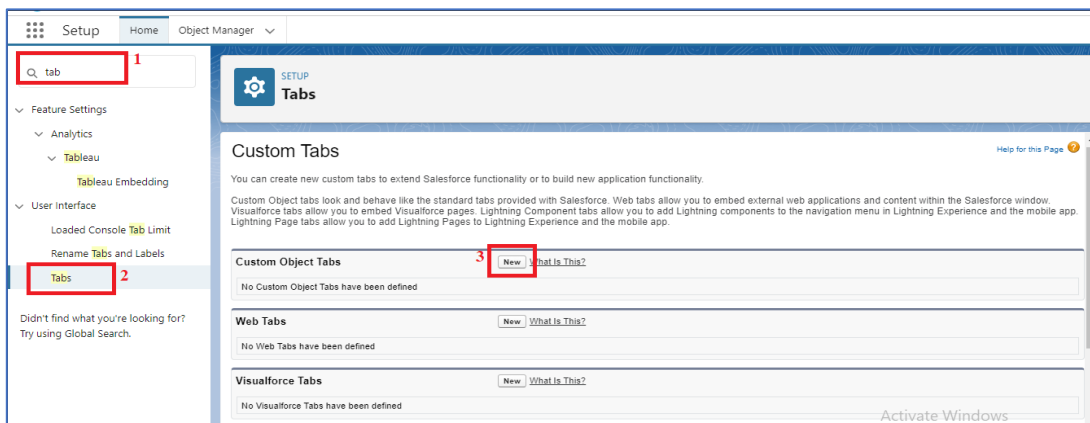
A screenshot of the 'Custom Object Tab Information' form in Salesforce. The form has a light blue header. Below the header, there are three rows of information: 'Tab Label' with the value 'Ticket Fares', 'Object' with the value 'Ticket Fare', and 'Tab Style' with a red 'Bell' icon. Below this, there is a section for 'Optional' settings, including 'Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.' and a dropdown menu for 'Splash Page Custom Link' set to '--None--'. There is a text area for 'Enter a short description.' with the label 'Description'. At the bottom of the form, there are 'Save' and 'Cancel' buttons.

6. Make sure that the Append tab to user's existing personal customizations is checked
7. Click on Save.

# A CRM Application For Public Transport Management

## Activity 5: Creating Tabs for Employee Object

1. Go to the setup page >> type Tabs in Quick Find bar
2. Click on tabs
3. Click on New (under custom object tab).



4. Select Object(Ticket Fare ) >> Select the tab style
5. Click on Next >>(Add to profiles page) keep it as default >>Click on Next (Add to Custom App) uncheck the include tab .

**Custom Object Tab Information**

|           |                                                                                                                                                                                 |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tab Label | Employees                                                                                                                                                                       |
| Object    | Employee                                                                                                                                                                        |
| Tab Style |  Compass  |

(Optional) Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.

Splash Page Custom Link --None--

Enter a short description.

Description

Save Cancel

6. Make sure that the Append tab to user's existing personal customizations is checked
7. Click on Save

# A CRM Application For Public Transport Management

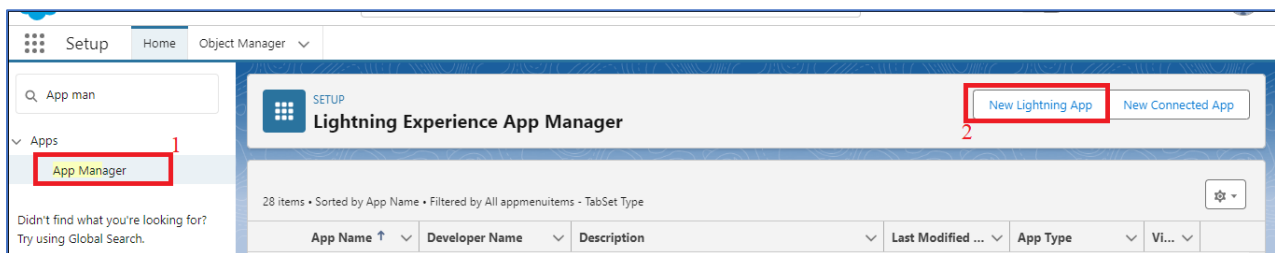
## Milestone 4

### The Lightning App

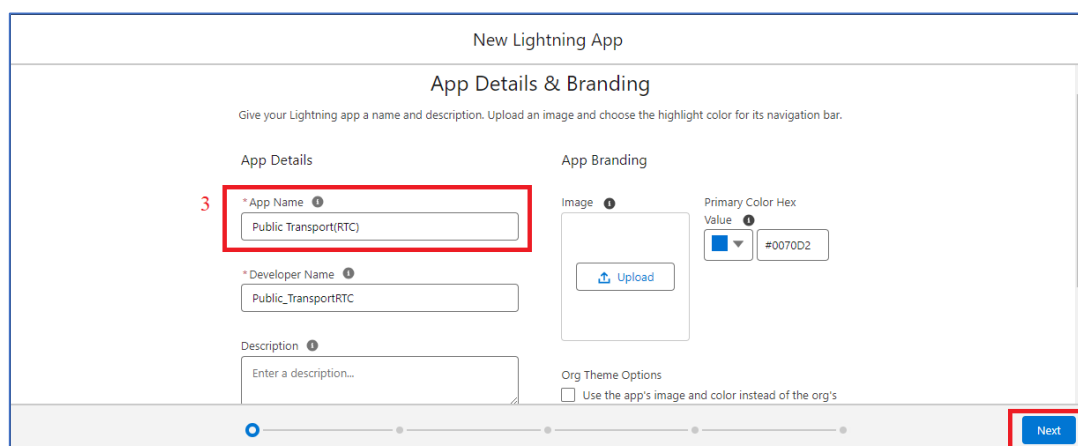
A Lightning App in Salesforce refers to an application built using the Lightning framework, which is a modern user interface framework for developing dynamic web applications for mobile and desktop devices. Lightning apps provide a more responsive and interactive user experience compared to traditional Visualforce pages.

#### Activity 1: Create a Lightning App for Banquet Hall Booking

1. From Setup, enter App Manager in the Quick Find and select App Manager.
2. Click New Lightning App.



3. Enter Public Transport as the App Name, then click next



4. Under App Options, leave the default selections and click next.
5. Under Utility Items, leave as is and click Next.
6. From Available Items, select Bus Stations, Buses, Trips, Ticket Fares, Employees, Reports, and Dashboards and move them to Selected Item and Click Next.

# A CRM Application For Public Transport Management

New Lightning App

### Navigation Items

Choose the items to include in the app, and arrange the order in which they appear. Users can personalize the navigation to add or move items, but users can't remove or rename the items that you add. Some navigation items are available only for phone or only for desktop. These items are dropped from the navigation bar when the app is viewed in a format that the item doesn't support.

Available Items

Create

Dash

Dashboards

Selected Items

- Bus stations
- Buses
- Employees
- Trips
- Ticket Fares
- Reports

Back

Next

7. From Available Profiles, select System Administrator and move it to Selected Profiles.
8. Click Save & Finish

New Lightning App

### User Profiles

Choose the user profiles that can access this app.

Available Profiles

system

System Administrator

Selected Profiles

Back

Save & Finish

# A CRM Application For Public Transport Management

## Milestone 5 Fields

| Object             | Field Name                     | Data Type                                      |
|--------------------|--------------------------------|------------------------------------------------|
| <b>Bus Station</b> | Bus Station Name(Standard)     | Text                                           |
|                    | Bus Stop Category              | Picklist(Unmanaged Bus Stop, Managed Bus Stop) |
|                    | Last Updated                   | Formula(Date)                                  |
|                    | Amenities                      | Picklist(Multi-select)                         |
|                    | City                           | Text(40)                                       |
|                    | Street                         | TextArea                                       |
|                    | State/Province                 | Text(25)                                       |
|                    | Zip/PostalCode                 | Text(10)                                       |
|                    | Shelter Available              | Checkbox                                       |
|                    | Bench                          | Checkbox                                       |
| <b>Bus</b>         | Bus Registration No (Standard) | Text                                           |
|                    | Bus Station Name               | Lookup(Bus Station)                            |
|                    | Capacity                       | Number(4,0)                                    |
|                    | Category                       | Picklist                                       |

## A CRM Application For Public Transport Management

|             | Model                 | Picklist(Dependent on Category) |
|-------------|-----------------------|---------------------------------|
| <b>Trip</b> | Trip No(Standard)     | Text                            |
|             | Trip Date             | Date                            |
|             | Bus No                | Lookup(Bus)                     |
|             | Route Name            | Lookup(Ticket Fare)             |
|             | Arrival Time          | Picklist                        |
|             | Departure Time        | Picklist                        |
|             | Bus Starting Terminal | Text                            |
|             | Destination Terminal  | Text                            |
|             | Driver Id             | Lookup(Employee)                |
|             | Driver                | Formula                         |
|             | Conductor Id          | Lookup(Employee)                |
|             | Conductor             | Formula                         |
|             | Estimated Travel Time | Number                          |
|             | Frequency Per Day     | Number(2,0)                     |
|             | No. of Stops          | Number(2,0)                     |
|             | Passenger Count       | Number(4,0)                     |
|             | Ticket Fare           | Currency(16,2)                  |
|             | Total Amount          | Formula                         |

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|                    |                       |                     |
|--------------------|-----------------------|---------------------|
| <b>Ticket Fare</b> | Route Name(Standard)  | Text                |
|                    | Bus Model             | Picklist            |
|                    | Ticket Fare           | Currency(10,2)      |
| <b>Employee</b>    | Employee Id(Standard) | Text                |
|                    | Bus Station Name      | Lookup(Bus Station) |
|                    | Employee Name         | Text                |
|                    | Role                  | Picklist            |
|                    | Date of Birth         | Date                |
|                    | Age                   | Formula(Number)     |
|                    | Work Place            | Text                |
|                    | Salary                | Currency(18,0)      |
|                    | Phone                 | Phone               |
|                    | Date of Joining       | Date                |
|                    | Date of Retirement    | Formula(Date)       |
|                    | Experience            | Formula(Number)     |
|                    | Street                | TextArea            |
|                    | City                  | Text                |
|                    | State/Province        | Text                |
|                    | Country               | Text                |

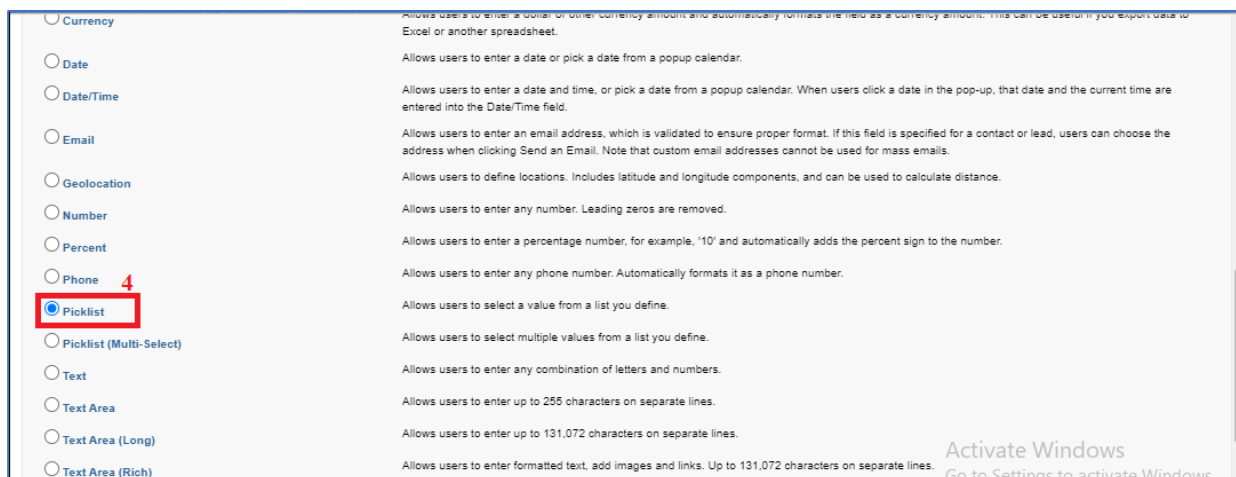
# A CRM Application For Public Transport Management

|  |                |      |
|--|----------------|------|
|  | Zip/PostalCode | Text |
|--|----------------|------|

## Activity 1: Creating a Role Picklist Field in Employee Object

To create fields in an object:

1. Go to setup >> click on Object Manager >> type object name(Employee) in quick find box>> click on the Employee Object.
2. Now click on "Fields & Relationships" .
3. Click on New.
4. Select Data type as "Picklist" and click Next.



5. Enter Field Label as "Role".
6. In values select "Enter values, with each value separated by a new line" and enter values as shown below.

Administrative Assistant

Cleaner

Conductor

Customer Service Representative

Driver



# A CRM Application For Public Transport Management

Safety Officer

Station Manager

Supervisor

Ticket Inspector

7. Select Display values alphabetically, not in the order entered .

The screenshot shows a configuration window titled "Step 2. Enter the details" for a field named "Role". The field label is "Role". Under the "Values" section, the option "Enter values, with each value separated by a new line" is selected. A list of values is displayed: Station Manager, Administrative Assistant, Customer Service Representative, Safety Officer, Supervisor, and Cleaner. The option "Display values alphabetically, not in the order entered" is checked. The "Field Name" is "Role". The "Description" field is empty. Navigation buttons "Previous", "Next", and "Cancel" are at the top right. A watermark "Activate Windows" is visible.

8. Select Required, Always require a value in this field in order to save a record .

9. Click on Next, Next and Save.

The screenshot shows the "Required" section of the configuration window. The "Required" checkbox is checked, and the text "Always require a value in this field in order to save a record" is displayed. The "Auto add to custom report type" checkbox is also checked, with the text "Add this field to existing custom report types that contain this entity". The "Default Value" section shows a "Show Formula Editor" button. A watermark "Activate Windows" is visible. Navigation buttons "Previous", "Next", and "Cancel" are at the bottom right.

# A CRM Application For Public Transport Management

## Activity 2: Creating a Trip Date Field in Trip object

To create fields in an object:

1. Go to setup >> click on Object Manager >> type object name(Trip) in quick find box>> click on the Trip object.
2. Now click on "Fields & Relationships"
3. Click on New.
4. Select Data type as "Date" and click Next.

This screenshot shows the 'New Field' dialog in Salesforce. The 'Date' radio button is selected and highlighted with a red box and the number '4'. Other options include Master-Detail Relationship, External Lookup Relationship, Checkbox, Currency, Date/Time, Email, Geolocation, Number, and Percent. Each option has a brief description of its functionality.

5. Enter Field Label as " Trip Date".
6. Select Required, Always require a value in this field in order to save a record .
7. Click on Next, Next and Save.

This screenshot shows the 'Field Properties' dialog for the 'Trip Date' field. The 'Field Label' is 'Trip Date' and the 'Data Type' is 'Date'. The 'Required' checkbox is checked, and the text 'Always require a value in this field in order to save a record' is displayed. The 'Field Name' is 'Trip\_Date'. The 'Compliance Categorization' section shows 'Available' categories: PII, HIPAA, GDPR, and PCI. The 'General Options' section at the bottom has a red box around the 'Required' checkbox and its description.

# A CRM Application For Public Transport Management

## Activity 3: Creating a Number Field in Bus object

To create fields in an object:

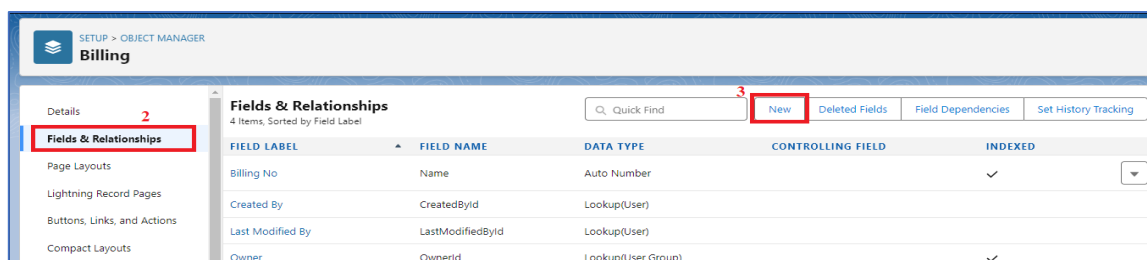
1. Go to setup >> click on Object Manager >> type object name(Bus) in quick find box >> click on the Bus object.
2. Now click on "Fields & Relationships"
3. Click on New.
4. Select Data type as "Number" and click Next.
5. Enter Field Label as " Capacity".
6. Length - 4, Decimal Places - 0.
7. Select Required, Always require a value in this field in order to save a record .
8. Click on Next, Next and Save.

## Activity 4 : Creating Lookup Relationship

A Lookup relationship is a type of relationship in Salesforce that connects two objects together based on a field known as the Lookup field. It establishes a relationship between a child object and a parent object, allowing the child object to reference the parent object.

To Create a relationship from Employee to Bus Station .

1. Go to the Setup page >> click on Object manager >> type object name(Employee) in the quick find bar >> click on the Employee object.
2. Click on fields & relationship
3. Click on New.



4. Select "Lookup relationship" as data type and click Next.

# A CRM Application For Public Transport Management

**Data Type**

Select one of the data types below.

- ☐ None Selected
- ☐ Auto Number
- ☐ Formula
- ☐ Roll-Up Summary
- ☒ **Lookup Relationship**
- ☐ Master-Detail Relationship

Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where:

- The relationship field is required on all detail records.
- The ownership and sharing of a detail record are determined by the master record.
- When a user deletes the master record, all detail records are deleted.
- You can create rollup summary fields on the master record to summarize the detail records.

5. Select the related object " Bus Station".

Employee  
New Relationship

Step 2. Choose the related object

Select the other object to which this object is related.

Related To: **Bus Station**

Previous Next Cancel

6. Click on Next

7. Give Field Label as "Bus Station Name" .

Employee  
New Relationship

Step 3. Enter the label and name for the lookup field

Field Label: **Bus Station Name**

Field Name: **Bus\_Station\_Name**

Description:

Help Text:

Child Relationship Name: **Employees**

Required: ☐ Always require a value in this field in order to save a record

What to do if the lookup record is deleted: ☒ Clear the value of this field. You can't choose this option if you make this field required.

Previous Next Cancel

8. Click on Next , Next, Next , Save.

# A CRM Application For Public Transport Management

## Activity 5: Creating a Checkbox Field in Bus Station object

To create fields in an object:

1. Go to setup >> click on Object Manager >> type object name(Bus Station) in quick find box>> click on the Bus Station object.
2. Now click on "Fields & Relationships"
3. Click on New.
4. Select Data type as "Checkbox" and click Next.
5. Enter Field Label as " Shelter available".

Bus Station  
New Custom Field

Step 2. Enter the details

Field Label: Shelter Available

Default Value: ☐ Checked ☒ Unchecked

Field Name: Shelter\_Available

Description: Indicates whether the bus stop has a shelter or covered waiting area.

Help Text:

Auto add to custom report type: ☒ Add this field to existing custom report types that contain this entity

Previous Next Cancel

Activate Windows  
Go to Settings to activate Windows.

6. Select Default value : Unchecked .
7. Click on Next, Next and Save.

## Activity 6: Creating a Phone Field in Employee object

To create fields in an object:

1. Go to setup >> click on Object Manager >> type object name(Employee) in quick find box>> click on the Employee object.
2. Now click on "Fields & Relationships"
3. Click on New.
4. Select Data type as "Phone" and click Next.
5. Enter Field Label as " Phone No".

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Employee New Custom Field

Step 2 of 4: Enter the details

Field Label: Phone

Field Name: Phone

Description:

Help Text:

Required: ☐ Always require a value in this field in order to save a record

Auto add to custom report type: ☒ Add this field to existing custom report types that contain this entity

Default Value: Show Formula Editor

Previous Next Cancel

Activate Windows  
Go to Settings to activate Windows.

6. Click on Next, Next and Save.

## Activity 7: Creating a Last Updated Formula Field in Bus Station object

To create fields in an object:

1. Go to setup >> click on Object Manager >> type object name(Bus Station) in quick find box >> click on the Bus Station object.
2. Now click on "Fields & Relationships"
3. Click on New.
4. Select Data type as "Formula" and click Next.
5. Enter field label Last Updated

Field Information

Field Label: Last Updated

Field Name: Last\_Updated

Description:

Help Text:

6. Select formula return type Date, Click Next
7. Create and insert Advance formula: TODAY()

# A CRM Application For Public Transport Management



The screenshot shows a formula editor interface with two tabs: "Simple Formula" and "Advanced Formula". The "Simple Formula" tab is active. Below the tabs, there are two buttons: "Insert Field" on the left and "Insert Operator" with a dropdown arrow on the right. The main area of the editor contains the text "Last Updated (Date) =" followed by a horizontal line, and below that, the text "TODAY ()".

8. Click Next, Next, then Save.

# A CRM Application For Public Transport Management

## Activity 8: Creating a Age Formula Field in Employee object

1. Go to setup >> click on Object Manager >> type object name(Employee) in quick find box >> click on the Employee object.
2. Now click on "Fields & Relationships"
3. Click on New.
4. Select Data type as "Formula" and click Next.
5. Enter field label Age

Step 2. Choose output type

Field Label: Age

Field Name: Age

Auto add to custom report type: ☒ Add this field to existing custom report types that contain this entity

Formula Return Type

☐ None Selected

☐ Checkbox

☐ Currency

☐ Date

☐ Date/Time

☒ Number

☐ Percent

Select one of the data types below.

Calculate a boolean value.  
Example: `TODAY() > CloseDate`

Calculate a dollar or other currency amount and automatically format the field as a currency amount.  
Example: `Gross Margin = Amount - Cost__c`

Calculate a date, for example, by adding or subtracting days to other dates.  
Example: `Reminder Date = CloseDate - 7`

Calculate a date/time, for example, by adding a number of hours or days to another date/time.  
Example: `Next = NOW() + 1`

Calculate a numeric value.  
Example: `Fahrenheit = 1.8 * Celsius__c + 32`

Activate Windows  
Go to Settings to activate Windows.

6. Select formula return type Number, Click Next
7. Create and insert Advance formula: `YEAR( TODAY() ) - YEAR( Date_of_Birth__c )`

Employee  
New Custom Field

Step 3. Enter formula

Enter your formula and click Check Syntax to check for errors. Click the Advanced Formula subtab to use additional fields, operators, and functions.

Example: `Fahrenheit = 1.8 * Celsius__c + 32` [More Examples...](#)

Age (Number) =  
`YEAR( TODAY() ) - YEAR( Date_of_Birth__c )`

Functions  
-- All Function Categories --  
ABS  
ACOS  
ADDMONTHS  
AND  
ASCII  
ASIN  
ASIN

Activate Windows  
Go to Settings to activate Windows.

8. Click Next, Next, then Save.



# A CRM Application For Public Transport Management

## Activity 9: Creating a Date of Retirement Formula Field in Employee object

To create fields in an object:

1. Go to setup >> click on Object Manager >> type object name(Employee) in quick find box >> click on the Employee object.
2. Now click on "Fields & Relationships"
3. Click on New.
4. Select Data type as "Formula" and click Next.
5. Enter field label Date of Retirement
6. Select formula return type Date, Click Next
7. Create and insert Advance formula: `DATE(YEAR( Date_of_Birth__c ) + 55, MONTH(Date_of_Birth__c) , DAY(Date_of_Birth__c) )`
8. Click Next, Next, then Save.

The screenshot shows the 'Employee Custom Field' configuration page for a field named 'Date of Retirement'. The page has a header with the field name and a 'Back to Employee' link. Below the header is a 'Custom Field Definition Detail' section with buttons for 'Edit', 'Set Field-Level Security', 'View Field Accessibility', and 'Where is this used?'. The main content is divided into two sections: 'Field Information' and 'Formula Options'. The 'Field Information' section contains a table with the following details:

|                           |                                      |             |                                      |
|---------------------------|--------------------------------------|-------------|--------------------------------------|
| Field Label               | Date of Retirement                   | Object Name | Employee                             |
| Field Name                | Date_of_Retirement                   |             |                                      |
| API Name                  | Date_of_Retirement__c                |             |                                      |
| Description               |                                      |             |                                      |
| Help Text                 |                                      |             |                                      |
| Data Owner                |                                      |             |                                      |
| Field Usage               |                                      |             |                                      |
| Data Sensitivity Level    |                                      |             |                                      |
| Compliance Categorization |                                      |             |                                      |
| Created By                | sanjay.polisetty, 6/3/2025, 12:22 AM | Modified By | sanjay.polisetty, 6/3/2025, 12:22 AM |

The 'Formula Options' section contains a table with the following details:

|           |                                                                                       |
|-----------|---------------------------------------------------------------------------------------|
| Data Type | Formula                                                                               |
| Formula   | DATE(YEAR( Date_of_Birth__c ) + 55, MONTH(Date_of_Birth__c) , DAY(Date_of_Birth__c) ) |

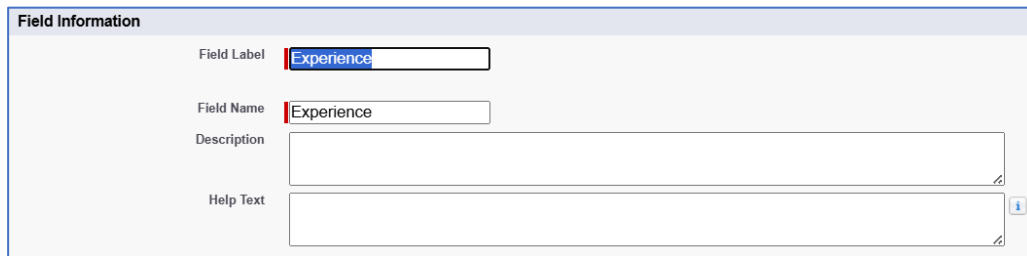
## Activity 10: Creating Experience Formula Field in Employee object

To create fields in an object:

1. Go to setup >> click on Object Manager type object name(Employee) in quick find box >> click on the Employee object.
2. Now click on "Fields & Relationships"

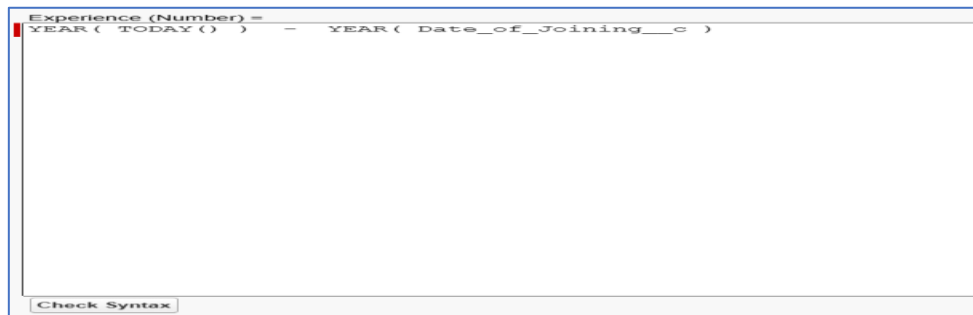
# A CRM Application For Public Transport Management

3. Click on New.
4. Select Data type as "Formula" and click Next.
5. Enter field label Experience



The 'Field Information' form shows the configuration for a new field. The 'Field Label' is 'Experience', the 'Field Name' is 'Experience', and the 'Description' and 'Help Text' fields are empty. The 'Field Type' is set to 'Formula'.

6. Select formula return type Number, Click Next
7. Create and insert Advance formula: YEAR( TODAY() ) - YEAR( Date\_of\_joining\_\_c )



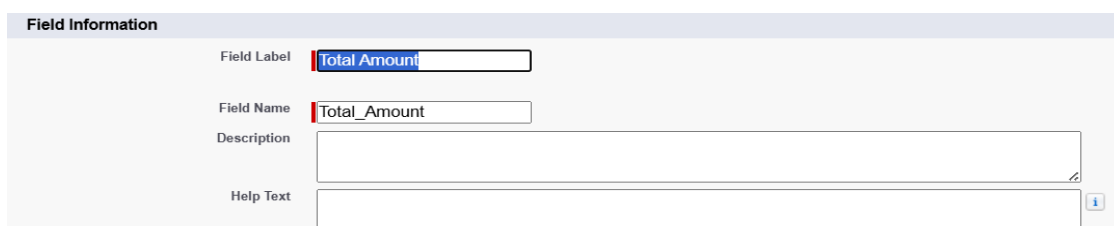
The formula editor shows the formula: `Experience (Number) = YEAR( TODAY() ) - YEAR( Date_of_Joining__c )`. A 'Check Syntax' button is at the bottom.

8. Click Next, Next, then Save.

## Activity 11: Creating a Total Amount Formula Field in Trip object

To create fields in an object:

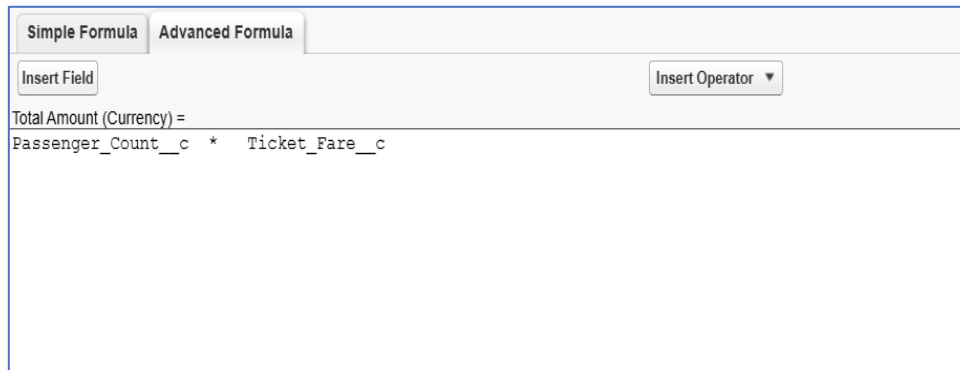
1. Go to setup >> click on Object Manager >> type object name(Trip) in quick find box >> click on the Trip object.
2. Now click on "Fields & Relationships"
3. Click on New.
4. Select Data type as "Formula" and click Next.
5. Enter field label Total Amount.



The 'Field Information' form shows the configuration for a new field. The 'Field Label' is 'Total Amount', the 'Field Name' is 'Total\_Amount', and the 'Description' and 'Help Text' fields are empty. The 'Field Type' is set to 'Formula'.

# A CRM Application For Public Transport Management

6. Select formula return type Currency, Click Next.
7. Create and insert Advance formula: Passenger\_Count\_\_c \* Ticket\_Fare\_\_c



Simple Formula   Advanced Formula

Insert Field   Insert Operator ▼

Total Amount (Currency) =

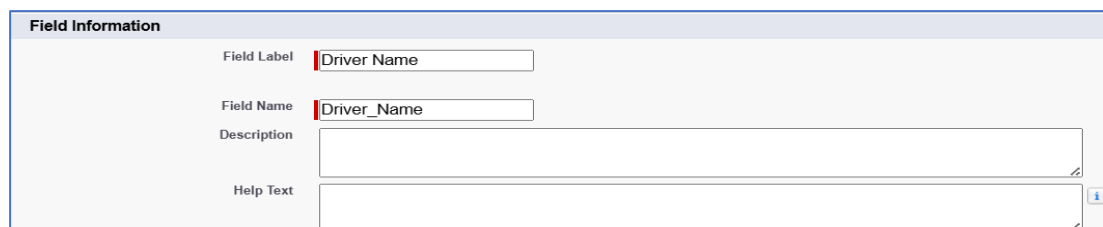
Passenger\_Count\_\_c \* Ticket\_Fare\_\_c

8. Click Next, Next, then Save.

## Activity 12: Creating a Driver Name Formula Field in Trip object

To create fields in an object:

1. Go to setup >> click on Object Manager >> type object name(Trip) in quick find box>> click on the Trip object.
2. Now click on "Fields & Relationships"
3. Click on New.
4. Select Data type as "Formula" and click Next.
5. Enter field label Driver Name.



Field Information

Field Label   Driver Name

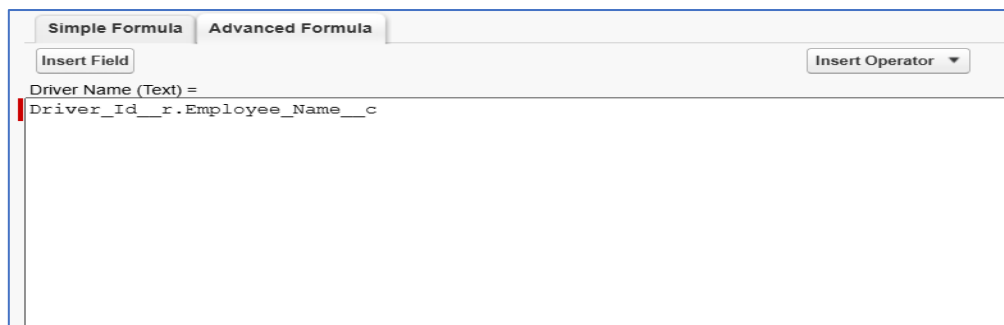
Field Name   Driver\_Name

Description

Help Text

6. Select formula return type Text, Click Next.
7. Create and insert Advance formula: Driver\_Id\_\_r.Employee\_Name\_\_c

# A CRM Application For Public Transport Management



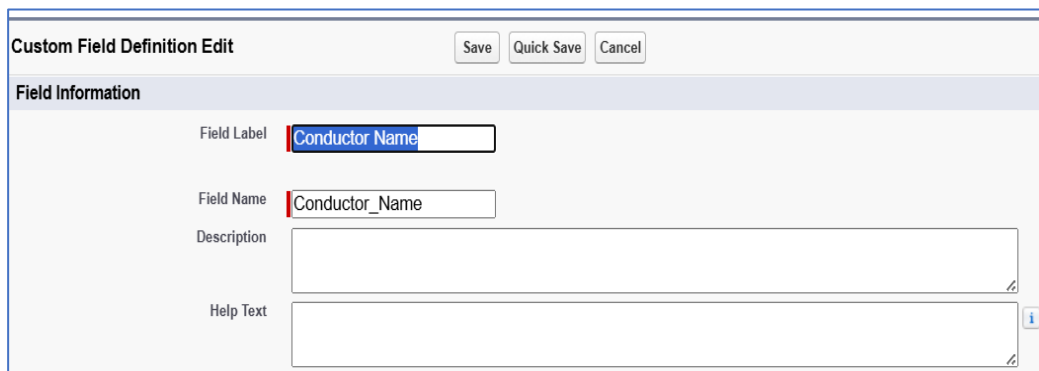
The screenshot shows a 'Simple Formula' editor window. At the top, there are two tabs: 'Simple Formula' (selected) and 'Advanced Formula'. Below the tabs, there are two buttons: 'Insert Field' on the left and 'Insert Operator' on the right. The main area contains the text 'Driver Name (Text) =' followed by a red vertical bar and the formula 'Driver\_Id\_\_r.Employee\_Name\_\_c'.

8. Click Next, Next, then Save.

## Activity 13: Creating a Conductor Name Formula Field in Trip object

To create fields in an object:

1. Go to setup >> click on Object Manager >> type object name(Trip) in quick find box >> click on the Trip object.
2. Now click on "Fields & Relationships"
3. Click on New.
4. Select Data type as "Formula" and click Next.
5. Enter field label Conductor Name.



The screenshot shows the 'Custom Field Definition Edit' window. At the top, there are three buttons: 'Save', 'Quick Save', and 'Cancel'. Below the buttons, there is a section titled 'Field Information'. This section contains four fields: 'Field Label' with the value 'Conductor Name', 'Field Name' with the value 'Conductor\_Name', 'Description' (empty), and 'Help Text' (empty). There is an information icon (i) next to the 'Help Text' field.

6. Select formula return type Text, Click Next.
7. Create and insert Advance formula: Conductor\_Id\_\_r.Employee\_Name\_\_c



The screenshot shows a 'Simple Formula' editor window. At the top, there are two tabs: 'Simple Formula' (selected) and 'Advanced Formula'. Below the tabs, there are two buttons: 'Insert Field' on the left and 'Insert Operator' on the right. The main area contains the text 'Conductor Name (Text) =' followed by a red vertical bar and the formula 'Conductor\_Id\_\_r.Employee\_Name\_\_c'.

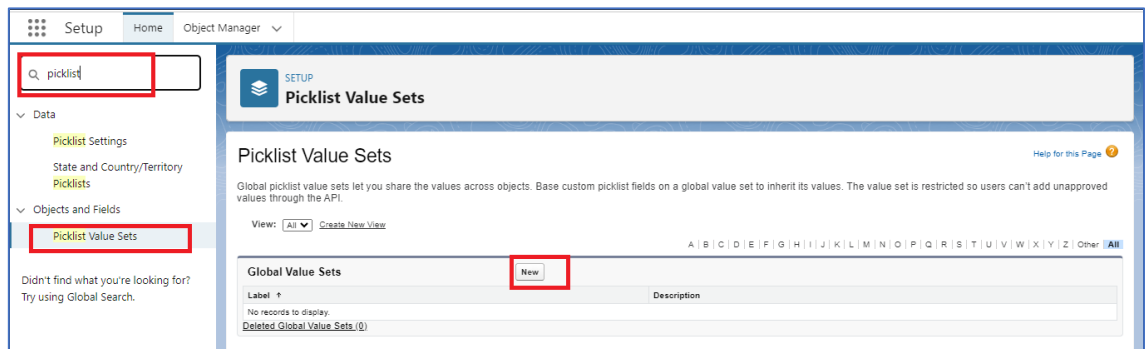
# A CRM Application For Public Transport Management

8. Click Next, Next, then Save.

## Activity 14: Create a Global Value Set

1.

1. From the Setup menu, enter "Picklist Value Sets" in the Quick Find box and select it.
2. Click on "New" to create a new global value set.



3. Enter the label Bus Time.
4. In values select "Enter values, with each value separated by a new line" and enter values as shown below.

6:00 AM

7:00 AM

8:00 AM

9:00 AM

10:00 AM

11:00 AM

12:00 PM

1:00 PM

2:00 PM

3:00 PM

4:00 PM

# A CRM Application For Public Transport Management

5:00 PM

6:00 PM

7:00 PM

8:00 PM

9:00 PM

10:00 PM

11:00 Pm

Global Value Set Edit

Save Save & New Cancel

Information

Label Bus Time

Name Bus\_Time

Description Time to select for Departure or Arrival time of Bus

Values

Enter values, with each value separated by a new line

9:00 AM  
10:00 AM  
11:00 AM  
12:00 PM  
1:00 PM  
2:00 PM  
3:00 PM  
4:00 PM  
5:00 PM  
6:00 PM  
7:00 PM  
8:00 PM  
9:00 PM  
10:00 PM

Activate Windows  
Go to Settings to activate Windows.

5. Click "Save" to create the global value set

## Activity 15: Create a Global Value Set for bus model

1. From the Setup menu, enter "Picklist Value Sets" in the Quick Find box and select it.
2. Click on "New" to create a new global value set.

Setup Home Object Manager

Q picklist

▼ Data

Picklist Settings

State and Country/Territory

Picklists

▼ Objects and Fields

Picklist Value Sets

Didn't find what you're looking for?  
Try using Global Search.

Global Value Sets

New

Global picklist value sets let you share the values across objects. Base custom picklist fields on a global value set to inherit its values. The value set is restricted so users can't add unapproved values through the API.

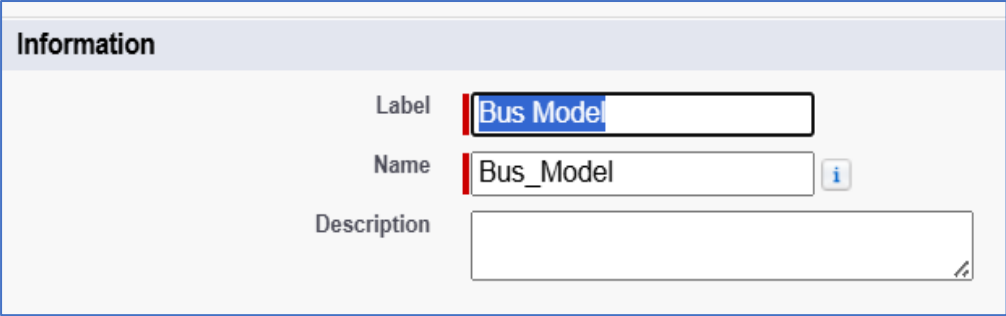
View: All Create New View

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Other All

| Global Value Sets             |             |
|-------------------------------|-------------|
| Label                         | Description |
| No records to display.        |             |
| Deleted Global Value Sets (0) |             |

3. Enter the label Bus Model.

# A CRM Application For Public Transport Management

A screenshot of a web application form titled 'Information'. It contains three fields: 'Label' with the value 'Bus Model', 'Name' with the value 'Bus\_Model' and an information icon, and 'Description' which is empty.

4. In values select "Enter values, with each value separated by a new line" and enter values as shown below

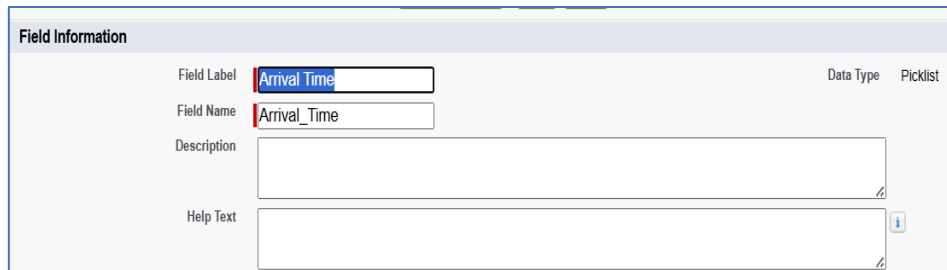
Regular  
Metro  
A/C  
Express  
Deluxe  
Super Deluxe  
Semi Sleeper  
Sleeper

## Activity 16: Creating a Picklist Field using global picklist value set in Trip Object

To create fields in an object:

1. Go to setup >> click on Object Manager >> type object name(Employee) in quick find box>> click on the Employee Object.
2. Now click on "Fields & Relationships" .
3. Click on New.
4. Select Data type as "Picklist" and click Next.
5. Enter Field Label as "Arrival Time".

# A CRM Application For Public Transport Management



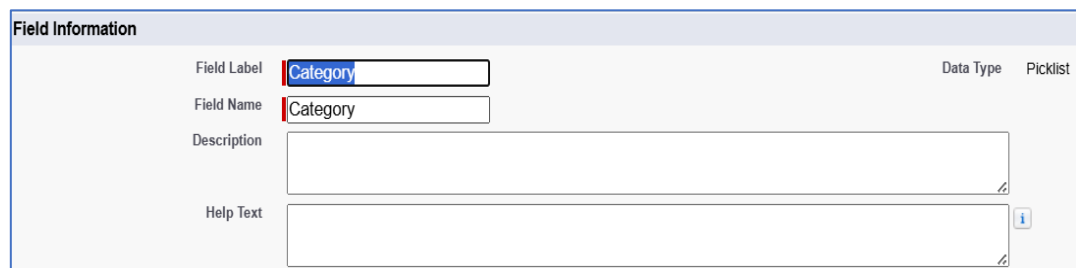
The screenshot shows a 'Field Information' form. The 'Field Label' is 'Arrival Time' and the 'Field Name' is 'Arrival\_Time'. The 'Data Type' is 'Picklist'. The 'Description' and 'Help Text' fields are empty. There is a small 'i' icon in the bottom right corner of the 'Help Text' field.

6. Select "Use global picklist value set" and choose the global value set "Bus Time".
7. Click on Next, Next and Save.

## Activity 17: Creating a Controlled and Dependent Picklists in Bus object

Creating Controlling picklist field in Bus object:

1. Go to setup >> click on Object Manager >> type object name(Bus) in quick find box >> click on the Bus Object.
2. Now click on "Fields & Relationships" .
3. Click on New.
4. Select Data type as "Picklist" and click Next.
5. Enter Field Label as "Category".



The screenshot shows a 'Field Information' form. The 'Field Label' is 'Category' and the 'Field Name' is 'Category'. The 'Data Type' is 'Picklist'. The 'Description' and 'Help Text' fields are empty. There is a small 'i' icon in the bottom right corner of the 'Help Text' field.

6. In values select "Enter values, with each value separated by a new line" and enter values as shown below.  
Local  
Intercity  
InterState
7. Select Required, Always require a value in this field in order to save a record .



# A CRM Application For Public Transport Management

**General Options**

Required ☒ Always require a value in this field in order to save a record

Unique ☐ Do not allow duplicate values

External ID ☐ Set this field as the unique record identifier from an external system

AI Prediction ☐ Use this field to store AI prediction scores

Default Value [Show Formula Editor](#)

Use formula syntax: Enclose text and picklist value API names in double quotes : ("the\_text"), include numbers without quotes : (25), show percentages as decimals: (0.10), and express date calculations in the standard format: (Today() + 7). To reference a field from a Custom Metadata type record use: \$CustomMetadata.Type\_\_mdt.RecordAPIName.Field\_\_c

8. Click on Next, Next and Save.

Creating Dependent picklist field in Bus object:

1. Go to setup >> click on Object Manager >> type object name(Bus) in quick find box >> click on the Bus Object.
2. Now click on "Fields & Relationships" .
3. Click on New.
4. Select Data type as "Picklist" and click Next.
5. Enter Field Label as "Category".

**Field Information**

Field Label  Data Type Picklist

Field Name

Description

Help Text

6. Select "Use global picklist value set" and choose the global value set "Bus Time".
7. Select Required, Always require a value in this field in order to save a record .

**General Options**

Required ☒ Always require a value in this field in order to save a record

Unique ☐ Do not allow duplicate values

External ID ☐ Set this field as the unique record identifier from an external system

AI Prediction ☐ Use this field to store AI prediction scores

Default Value [Show Formula Editor](#)

Use formula syntax: Enclose text and picklist value API names in double quotes : ("the\_text"), include numbers without quotes : (25), show percentages as decimals: (0.10), and express date calculations in the standard format: (Today() + 7). To reference a field from a Custom Metadata type record use: \$CustomMetadata.Type\_\_mdt.RecordAPIName.Field\_\_c

8. Click on Next, Next and Save.

# A CRM Application For Public Transport Management

## Establish the Dependency

1. Go to setup >> click on Object Manager >> type object name(Bus) in quick find box >> click on the Bus Object
2. Now click on "Fields & Relationships" .
3. Now click on "Field Dependencies" .
4. Click on New.

Controlling Field: --None--

Dependent Field: --None--

Continue Cancel

Continue Cancel

5. Enter Controlling Field : Category
6. Enter Dependent field : Model
7. Click "Continue".
8. Matrix with the controlling field values on the top and the dependent field values on the side. Check the boxes to define which dependent picklist values should be available for each controlling picklist value.
9. Click Save

Click button to include or exclude selected values from the dependent picklist:

Include Values Exclude Values

Showing Columns: 1 - 3 (of 3) < Previous | Next > View All Go to

| Category: | Local        | Intercity    | Interstate   |
|-----------|--------------|--------------|--------------|
| Model:    | Regular      | Regular      | Regular      |
|           | Metro        | Metro        | Metro        |
|           | A/c          | A/c          | A/c          |
|           | Express      | Express      | Express      |
|           | Deluxe       | Deluxe       | Deluxe       |
|           | Super Deluxe | Super Deluxe | Super Deluxe |
|           | Semi Sleeper | Semi Sleeper | Semi Sleeper |
|           | Sleeper      | Sleeper      | Sleeper      |

Showing Columns: 1 - 3 (of 3) < Previous | Next > View All

Click button to include or exclude selected values from the dependent picklist:

Include Values Exclude Values

Save Cancel Preview

# A CRM Application For Public Transport Management

## Milestone 6

### Editing of Page Layouts

Page layouts in Salesforce control the layout and organization of fields, related lists, custom links, and other elements on a record detail or edit page. They are essential for managing how data is presented to users and can vary based on user roles and profiles.

#### Activity 1: To edit a Page Layout in Bus Station Object

1. Go to setup >> click on Object Manager >> type object name(Bus Station) in quick find box >> click on the Bus Station object >> Page Layouts .
2. Click on the Bus Station Layout.
3. Drag and Arrange the field as shown below.

The screenshot shows the Salesforce Page Layout Editor for the 'Bus Station' object. The interface includes a top toolbar with buttons for 'Save', 'Quick Save', 'Preview As...', 'Cancel', 'Undo', 'Redo', and 'Layout Properties'. The main area displays a grid of fields organized into sections:

- Information (Header visible on edit only):**
  - Bus Station Name (Sample Text)
  - Amenities (Sample Text)
  - Bus Stop Category (Sample Text)
  - Last Updated (10/06/2024) - This field is highlighted in blue.
  - Shelter Available (checked)
  - Bench (checked)
  - Owner (Sample Text)
- Address Information:**
  - Street (Sample Text)
  - City (Sample Text)
  - Location (Sample Text)
  - State/Province (Sample Text)
  - Zip/PostalCode (Sample Text)
- System Information (Header visible on edit only):**
  - Created By (Sample Text)
  - Last Modified By (Sample Text)

4. Click on field Last Updated >> click on settings >> select Read Only and save it.
5. Click on Save.

# A CRM Application For Public Transport Management

## Activity 2: To create a Page Layout in Bus Object

1. Go to setup >> click on Object Manager >> type object name(Bus) in quick find box >> click on the Bus object >> Page Layouts.
2. Click on the Bus Layout
3. Drag and Arrange the field as shown below

Save Quick Save Preview As... Cancel Undo Redo Layout Properties

**Bus Detail**

Standard Buttons: Edit Delete Clone Change Owner Change Record Type Printable View Sharing Sharing Hierarchy

Custom Buttons:

Information (Header visible on edit only)

|                  |             |       |             |
|------------------|-------------|-------|-------------|
| Bus Registration | No          | Owner | Sample Text |
| Bus Station Name | Sample Text |       |             |
| Category         | Sample Text |       |             |
| Model            | Sample Text |       |             |
| Capacity         | 4,072       |       |             |

System Information (Header visible on edit only)

|            |             |                  |             |
|------------|-------------|------------------|-------------|
| Created By | Sample Text | Last Modified By | Sample Text |
|------------|-------------|------------------|-------------|

Custom Links (Header visible on edit only)

4. Click Save

## Activity 3: To create a Page Layout in Employee Object

1. Go to setup >> click on Object Manager >> type object name(Employee) in quick find box >> click on the Employee object >> Page Layouts.
2. Click on the Employee Layout
3. Drag and Arrange the field as shown below

Save Quick Save Preview As... Cancel Undo Redo Layout Properties

**Employee Detail**

Standard Buttons: Edit Delete Clone Change Owner Change Record Type Printable View Sharing Sharing Hierarchy

Custom Buttons:

Information (Header visible on edit only)

|                  |             |                    |             |
|------------------|-------------|--------------------|-------------|
| Employee ID      | Sample Text | Work Place         | Sample Text |
| Employee Name    | Sample Text | Role               | Sample Text |
| Bus Station Name | Sample Text | Date of joining    | 10/06/2024  |
| Salary           | £123.45     | Date of Retirement | 10/06/2024  |
| Experience       | 794.17      | Owner              | Sample Text |

Personal Details (Header visible on edit only)

|               |                |     |        |
|---------------|----------------|-----|--------|
| Date of Birth | 10/06/2024     | Age | 35.190 |
| Phone         | 1 415 555 1212 |     |        |

Address (Header visible on edit only)

|        |             |                |             |
|--------|-------------|----------------|-------------|
| Street | Sample Text | State/Province | Sample Text |
| City   | Sample Text | Country        | Sample Text |
|        |             | Zip/PostalCode | Sample Text |

System Information (Header visible on edit only)

|            |             |                  |             |
|------------|-------------|------------------|-------------|
| Created By | Sample Text | Last Modified By | Sample Text |
|------------|-------------|------------------|-------------|

# A CRM Application For Public Transport Management

4. Click Save.

## Activity 4: To create a Page Layout in Trip Object

1. Go to setup >> click on Object Manager >> type object name(Trip) in quick find box >> click on the Trip object >> Page Layouts.
2. Click on the Trip Layout
3. Drag and Arrange the field as shown below

The screenshot shows the 'Layout Properties' window for the Trip object. It contains three sections of fields:

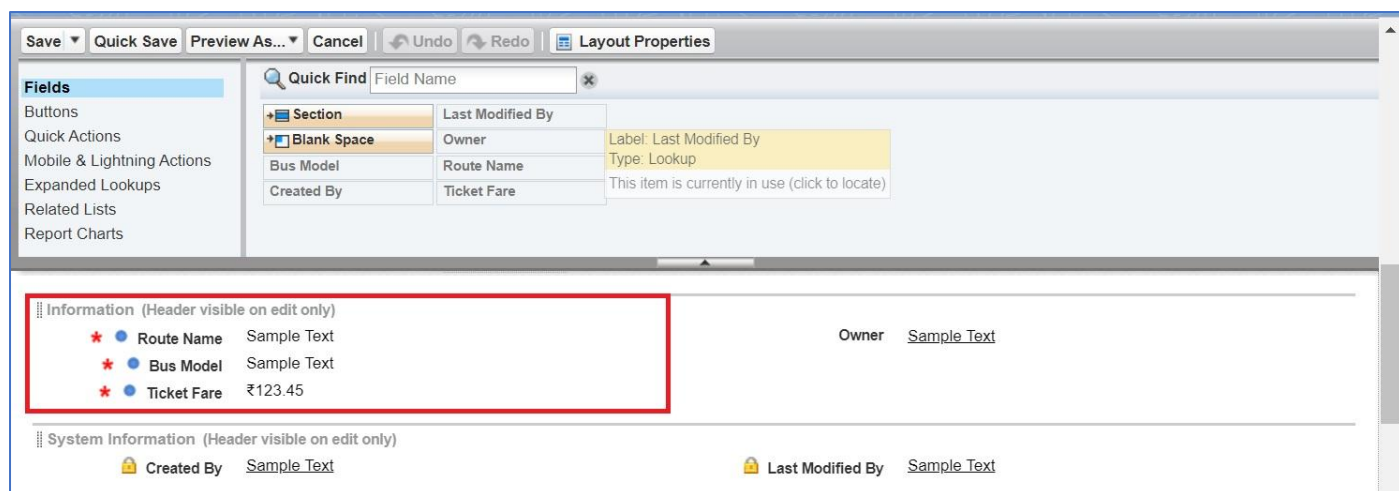
- Information (Header visible on edit only):**
  - Trip No: Sample Text
  - Trip Date: 10/06/2024
  - Bus No: Sample Text
  - Driver Id: Sample Text
  - Driver Name: Sample Text
  - Conductor Id: Sample Text
  - Conductor Name: Sample Text
  - Owner: Sample Text
- Bus Schedule:**
  - Route Name: Sample Text
  - Bus Starting Terminal: Sample Text
  - Departure Time: Sample Text
  - No. of Stops: 34
  - Estimated Travel Time: 995.77
  - Destination Terminal: Sample Text
  - Arrival Time: Sample Text
  - Frequency Per Day: 45
- Passenger Information:**
  - Passenger Count: 5,691
  - Ticket Fare: ₹123.45
  - Total Amount: ₹123.45

4. Click on field Ticket Fare >> click on settings >> select Read Only and save it.
5. Click Save.

## Activity 5: To create a Page Layout in Ticket Fare Object

1. Go to setup >> click on Object Manager >> type object name(Ticket Fare) in quick find box >> click on the Ticket Fare object >> Page Layouts.
2. Click on the Ticket Layout
3. Drag and Arrange the field as shown below

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Save Quick Save Preview As... Cancel Undo Redo Layout Properties

Quick Find Field Name

| Section     | Last Modified By |
|-------------|------------------|
| Blank Space | Owner            |
| Bus Model   | Route Name       |
| Created By  | Ticket Fare      |

Label: Last Modified By  
Type: Lookup  
This item is currently in use (click to locate)

Information (Header visible on edit only)

|             |             |
|-------------|-------------|
| Route Name  | Sample Text |
| Bus Model   | Sample Text |
| Ticket Fare | ₹123.45     |

Owner Sample Text

System Information (Header visible on edit only)

|                  |             |
|------------------|-------------|
| Created By       | Sample Text |
| Last Modified By | Sample Text |

4. Click Save.

# A CRM Application For Public Transport Management

## Milestone 7 Validation Rules

Validation rules in Salesforce are used to ensure data integrity by preventing users from entering invalid data into fields. They consist of a logical formula or expression that evaluates the data in one or more fields and returns true or false. If the rule returns true, an error message is displayed, and the record is not saved.

### Activity 1: To create a validation rule to a Employee Object

1. Go to setup >> click on Object Manager >> type object name(Employee) in quick find box >> click on the Employee object
2. Click on the validation rule >> click on New.
3. Enter the Rule name as "Employee\_Age\_Checking".
4. Select Active
5. Insert the Error Condition Formula as :

Age\_\_c < 21

The screenshot shows the Salesforce Object Manager interface for the 'Employee' object. The left sidebar contains navigation links: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area is titled 'Employee' and shows the configuration for a new validation rule. The 'Rule Name' field is set to 'Employee\_Age\_Checking' (marked with a red box and number 3). The 'Active' checkbox is checked (marked with a red box and number 4). The 'Description' field is empty. The 'Error Condition Formula' section shows an example formula: 'Discount\_Percent\_\_c > 0.30' with a note 'Display an error if Discount is more than 30%'. Below this, it states 'If this formula expression is true, display the text defined in the Error Message area'. The formula input field contains 'Age\_\_c < 21' (marked with a red box and number 5). To the right of the formula input is a 'Functions' dropdown menu showing a list of functions: ATAN, ATAN2, BEGINS, BLANKVALUE, BR, and CASE. A 'Quick Tips' box on the right contains a link to 'Operators & Functions'.

6. Enter the Error Message as "Employee Age Must be Greater than or equal to 21".
7. Select the Error location as Top of Page
8. Click Save.

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## Activity 2: To create a validation rule to a Bus Station on Employee Object

1. Go to setup >> click on Object Manager >> type object name(Employee) in quick find box >> click on the Banquet Hall object
2. Click on the validation rule >> click on New.
3. Enter the Rule name as "Employees\_only\_for\_Managed\_Bus\_stops".

Rule Name: Employees\_only\_for\_Managed\_Bus\_stops (3)

Active: ☒ (4)

Description:

**Error Condition Formula** I = Required Information

Example: Discount\_Percent\_\_c>0.30 [More Examples...](#)  
Display an error if Discount is more than 30%  
If this formula expression is **true**, display the text defined in the Error Message area

Insert Field Insert Operator (5)

IF( ISPICKVAL( Bus\_Station\_Name\_\_r.Bus\_Stop\_Category\_\_c , "UnManaged Bus Stop") , true, false)

Functions: -- All Function Categories --  
ABS  
ACOS  
ADDMONTHS  
AND  
ASCII  
ASIN  
Insert Selected Function  
ABS(number)

4. Select Active
5. Insert the Error Condition Formula as :  
IF( ISPICKVAL( Bus\_Station\_Name\_\_r.Bus\_Stop\_Category\_\_c , "UnManaged Bus Stop") , true, false)
6. Enter the Error Message as "The Employees must work for Managed Bus stops".
7. Select the Error location as Field and as Bus Station Name and click Save.

**Error Message**

Example: Discount percent cannot exceed 30%

This message will appear when Error Condition formula is **true** (6)

Error Message: The Employees must work for Managed Bus stops

This error message can either appear at the top of the page or below a specific field on the page

Error Location: ☐ Top of Page ☒ Field Bus Station Name (7)

Save Save & New Cancel



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## Activity 3: To create a validation rule to a Phone No on Employee Object

1. Go to setup >> click on Object Manager >> type object name(Employee) in quick find box>> click on the Employee object
2. Click on the validation rule >> click on New.
3. Enter the Rule name as "Phone\_Number\_Validation".
4. Select Active
5. Insert the Error Condition Formula as :  
IF( NOT(REGEX( Phone\_\_c , "[6-9]\\d{9}\$" ) ) , true, false)

The screenshot shows the Salesforce validation rule configuration page for 'Phone\_Number\_Validation'. The 'Rule Name' field is set to 'Phone\_Number\_Validation' and the 'Active' checkbox is checked. The 'Error Condition Formula' section contains the formula: `IF( NOT(REGEX( Phone__c , "[6-9]\\d{9}$" ) ) , true, false)`. The formula is entered in a text area with 'Insert Field' and 'Insert Operator' buttons above it. To the right, a 'Functions' list is visible, showing categories like 'All Function Categories' and specific functions like 'ABS', 'ACOS', 'ADDMONTHS', 'AND', 'ASCII', and 'ASIN'. A 'Quick Tips' box in the top right corner points to 'Operators & Functions'.

6. Enter the Error Message as "Phone no must be 10 digits and starts with 6 or 7 or 8or9".
7. Select the Error location as Top of Page and click Save.

## Activity 4: To create a validation rule to a Trip Object

1. Go to setup >> click on Object Manager >> type object name(Trip) in quick find box >> click on the Trip object
2. Click on the validation rule >> click on New.

# A CRM Application For Public Transport Management

3. Enter the Rule name as "Departure\_and\_Arrival\_Time\_Checking".
4. Select Active
5. Insert the Error Condition Formula as :

`TEXT( Departure_Time__c ) = TEXT( Arrival_Time__c )`

Rule Name:  3

Active: ☒ 4

Description:

**Error Condition Formula** ⓘ = Required Information

Example:  [More Examples...](#)  
Display an error if Discount is more than 30%  
If this formula expression is **true**, display the text defined in the Error Message area

Insert Field Insert Operator 5

`TEXT( Departure_Time__c ) = TEXT( Arrival_Time__c )`

Functions: -- All Function Categories --  
ABS  
ACOS  
ADDMONTHS  
AND  
ASCII  
ASIN

Quick Tips  
• [Operators & Functions](#)

6. Enter the Error Message as "The Departure Time and Arrival Time Should not be the same".
7. Select the Error location as Top of Page and click Save.

**Error Message**

Example:

This message will appear when Error Condition formula is true

Error Message:  6

This error message can either appear at the top of the page or below a specific field on the page

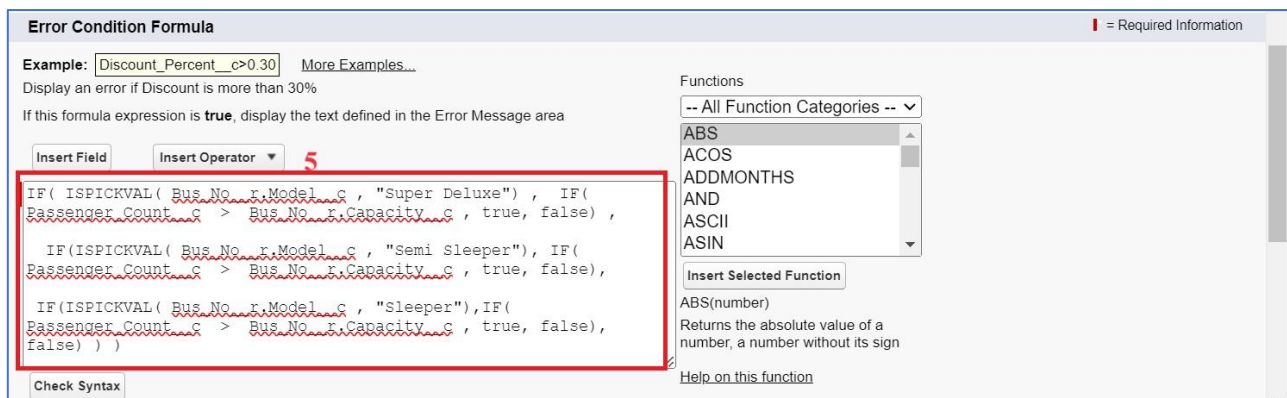
Error Location: ☐ Top of Page ☒ Field  ⓘ 7

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## Activity 5: To create a second validation rule to a Trip Object

1. Go to setup >> click on Object Manager >> type object name(Trip) in quick find box >> click on the Trip object
2. Click on the validation rule >> click on New.
3. Enter the Rule name as "Passenger\_Count\_Checking\_for\_Few\_Buses".
4. Select Active
5. Insert the Error Condition Formula as :

```
IF( ISPICKVAL( Bus_No__r.Model__c , "Super Deluxe") , IF( Passenger_Count__c > Bus_No__r.Capacity__c , true, false) ,  
IF(ISPICKVAL( Bus_No__r.Model__c , "Semi Sleeper"), IF( Passenger_Count__c > Bus_No__r.Capacity__c , true, false),  
IF(ISPICKVAL( Bus_No__r.Model__c , "Sleeper"),IF( Passenger_Count__c > Bus_No__r.Capacity__c , true, false), false) ) )
```



6. Enter the Error Message as "For Super Deluxe, Semi Sleeper and Sleeper Buses ,the Passenger Count must be less than or equal to the Capacity of the Bus".
7. Select the Error location as Field and as Passenger Count and click Save.

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## Activity 6: To create a validation rule to a Bus Object

1. Go to setup >> click on Object Manager >> type object name(Bus) in quick find box >> click on the Bus Hall object
2. Click on the validation rule >> click on New.
3. Enter the Rule name as "Bus\_Registration\_Number\_Validation".
4. Select Active
5. Insert the Error Condition Formula as :

IF( NOT( REGEX( Name , "^ [A-Z]{2} \s \d{2} \s [A-Z]{1,2} \s \d{4} \$" ) ), true, false)

Rule Name: Bus\_Registration\_Number\_Validation

Active: ☒

Description:

Error Condition Formula

Example: Discount\_Percent\_c>0.30 More Examples...

Display an error if Discount is more than 30%

If this formula expression is true, display the text defined in the Error Message area

Insert Field Insert Operator

IF( NOT( REGEX( Name , "^ [A-Z]{2} \s \d{2} \s [A-Z]{1,2} \s \d{4} \$" ) ), true, false)

Functions

-- All Function Categories --

ABS  
ACOS  
ADDMONTHS  
AND  
ASCII  
ASIN

Insert Selected Function

ABS(number)

Quick Tips

- Operators & Functions

6. Enter the Error Message as "The bus Registration Number must be in the format of 2 Capital Letters(State Code), space, 2 Numbers (District Code), space 1 or 2 Capital Letters(Series), space and 4 Numbers (Number).".
7. Select the Error location as Field and as Bus Registration No and click Save.

# A CRM Application For Public Transport Management

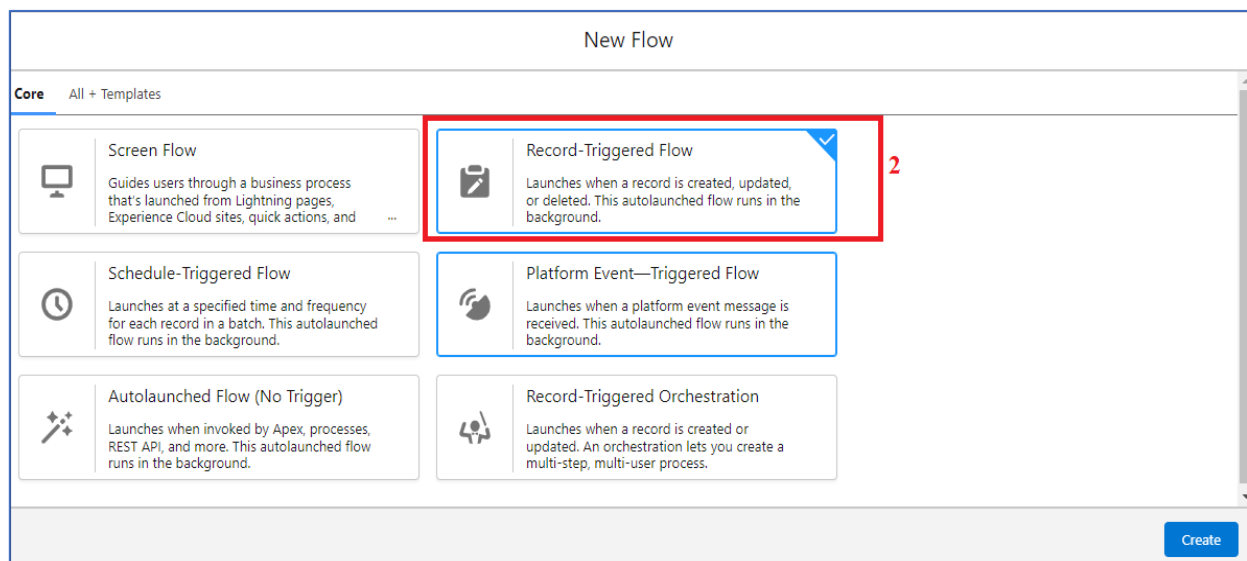
## Milestone 8 Flows

In Salesforce, a Flow is an automation tool that allows you to create complex business processes using a visual interface. Flows can be used to collect, update, delete, and create records, as well as to guide users through a series of screens to complete a process.

### Activity 1 : Create Flow to Fetch Ticket Fare for Bus.

Note: Please enter Route Name in Ticket Fare object as Hyderabad-Warangal(Express) , Hyderabad-Warangal(Deluxe), ...

1. Go to setup ?type Flow in quick find box ? Click on the Flow and Select the New Flow.
2. Select the record Triggered flow.Click on create.



3. Under Object select "Trip"
4. Select A record is created or updated

# A CRM Application For Public Transport Management

Configure Start

Select Object

Select the object whose records trigger the flow when they're created, updated, or deleted.

\* Object  
Trip

Configure Trigger

\* Trigger the Flow When:

☐ A record is created

☐ A record is updated

☒ A record is created or updated

☐ A record is deleted

5. Set Entry Conditions : None

6. Select Actions and Related Records and click on Don

Configure Start

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.

If you create a flow that's triggered when a record is updated, we recommend first defining entry conditions. Then select the **Only when a record is updated to meet the condition requirements** option for When to Run the Flow for Updated Records.

Condition Requirements  
None

\* Optimize the Flow for:

Fast Field Updates  
Update fields on the record that triggers the flow to run. This high-performance flow runs *before* the record is saved to the database.

Actions and Related Records  
Update any record and perform actions, like send an email. This more flexible flow runs *after* the record is saved to the database.

☐ Include a Run Asynchronously path to access an external system after the original transaction for the triggering record is successfully committed

Cancel Done

7. Under the record trigger flow click on the "+" icon and select Get Records.

8. Enter Label as " Fetching Route Ticket Fares ".

9. For Object select Ticket Fare

Find Salesforce records and store their field values in flow variables.

Label  
Fetching Route Ticket Fares

\* API Name  
Fetching\_Route\_Ticket\_Fares

Description  
Getting the records from Ticket Fares based on the bus model

Get Records of This Object

\* Object  
Ticket Fare

# A CRM Application For Public Transport Management

10. For Condition Requirements , select All Conditions are Met(AND)

For the first condition select as follows:

Field: Id

Operator: Equals

Value: {!\$Record.Route\_Name\_\_c}

Filter Ticket Fare Records

Condition Requirements

All Conditions Are Met (AND)

| Field | Operator | Value                 |
|-------|----------|-----------------------|
| Id    | Equals   | \$Record > Route Name |

+ Add Condition

11. For How many Records to store Select Only the First Record.

12. For How to Store Record Data select Choose fields and let Salesforce do the rest. Select

Field: Ticket\_Fare\_\_c. Click on Done.

How Many Records to Store

☒ Only the first record

☐ All records

How to Store Record Data

☐ Automatically store all fields

☒ Choose fields and let Salesforce do the rest

☐ Choose fields and assign variables (advanced)

Select Ticket Fare Fields to Store in Variable

| Field          |
|----------------|
| ID             |
| Ticket_Fare__c |

+ Add Field

Cancel Done

# A CRM Application For Public Transport Management

13. From the Toolbox drag and drop Decision element.

14. Enter the Decision label as "Bus Model Checking".

15. For Outcome Details:

- Label : Ticket Fare Checking
- Outcome API Name : Ticket\_Fare\_Checking
- Condition Requirements to Execute Outcome : All Conditions are Met (AND)
- Resource : {!Fetching\_Route\_Ticket\_Fares.Ticket\_Fare\_\_c}
- Operator : Is Null
- Value : {!\$GlobalConstant.False}

The screenshot displays the Salesforce Flow Builder interface for configuring a Decision element. The main configuration area is titled 'Bus Model Checking' and includes the following fields:

- \* Label:** Bus Model Checking
- \* API Name:** Bus\_Model\_Checking
- Description:** (Empty text area)
- Outcomes:** For each path the flow can take, create an outcome. For each outcome, specify the conditions that must be met for the flow to take that path.

The 'Outcomes' section is expanded, showing the configuration for the 'Ticket Fare Checking' outcome:

- OUTCOME ORDER:** 1
- \* Label:** Ticket Fare Checking
- \* Outcome API Name:** Ticket\_Fare\_Checking
- Condition Requirements to Execute Outcome:** All Conditions Are Met (AND)
- Resource:** {!Fetching\_Route\_Ticket\_Fares.Ticket\_Fare\_\_c}
- Operator:** Is Null
- Value:** {!\$GlobalConstant.False}

At the bottom of the configuration area, there is a '+ Add Condition' button. The bottom right corner of the interface shows 'Cancel' and 'Done' buttons.

16. Click Done

17. From the Toolbox drag and drop Update Records element and connect to Decision element for Ticket Fare Fetching Output.

18. Enter the label as "Updating Trip Object Ticket Fare Field".

19. How to Find Records to Update and Set Their Values : Use the trip record that triggered the flow

20. Set Filter Conditions : None -Always Update Record



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21. Field : Ticket\_Fare\_\_c

Value : {!Fetching\_Route\_Ticket\_Fares.Ticket\_Fare\_\_c}

And click Done

**\* How to Find Records to Update and Set Their Values**

- ☒ Use the trip record that triggered the flow
- ☐ Update records related to the trip record that triggered the flow
- ☐ Use the IDs and all field values from a record or record collection
- ☐ Specify conditions to identify records, and set fields individually

**Set Filter Conditions**

Condition Requirements to Update Record

None—Always Update Record

**Set Field Values for the Trip Record**

| Field          | Value                                         |
|----------------|-----------------------------------------------|
| Ticket_Fare__c | {!Fetching_Route_Ticket_Fares.Ticket_Fare__c} |

+ Add Field

Cancel Done

22. From the Toolbox drag and drop Custom Error Message element and connect to Default Outcome of Decision element..

23. Enter the label as "Route with Bus Model Does not Exists".

24. For Where to Show the Error Message: Select In a window on a record page

Error Message: There is no Record with the Route {!\$Record.Route\_Name\_\_r.Name} and Bus Model {!\$Record.Bus\_No\_\_r.Model\_\_c} in the Ticket Fares

**Set Error Message 1 Details**

**Where to Show the Error Message:**

- ☒ In a window on a record page
- ☐ As an inline error on a field

**Error Message**

\* Enter an error message with no more than 255 characters.

Resource Picker

Insert a resource...

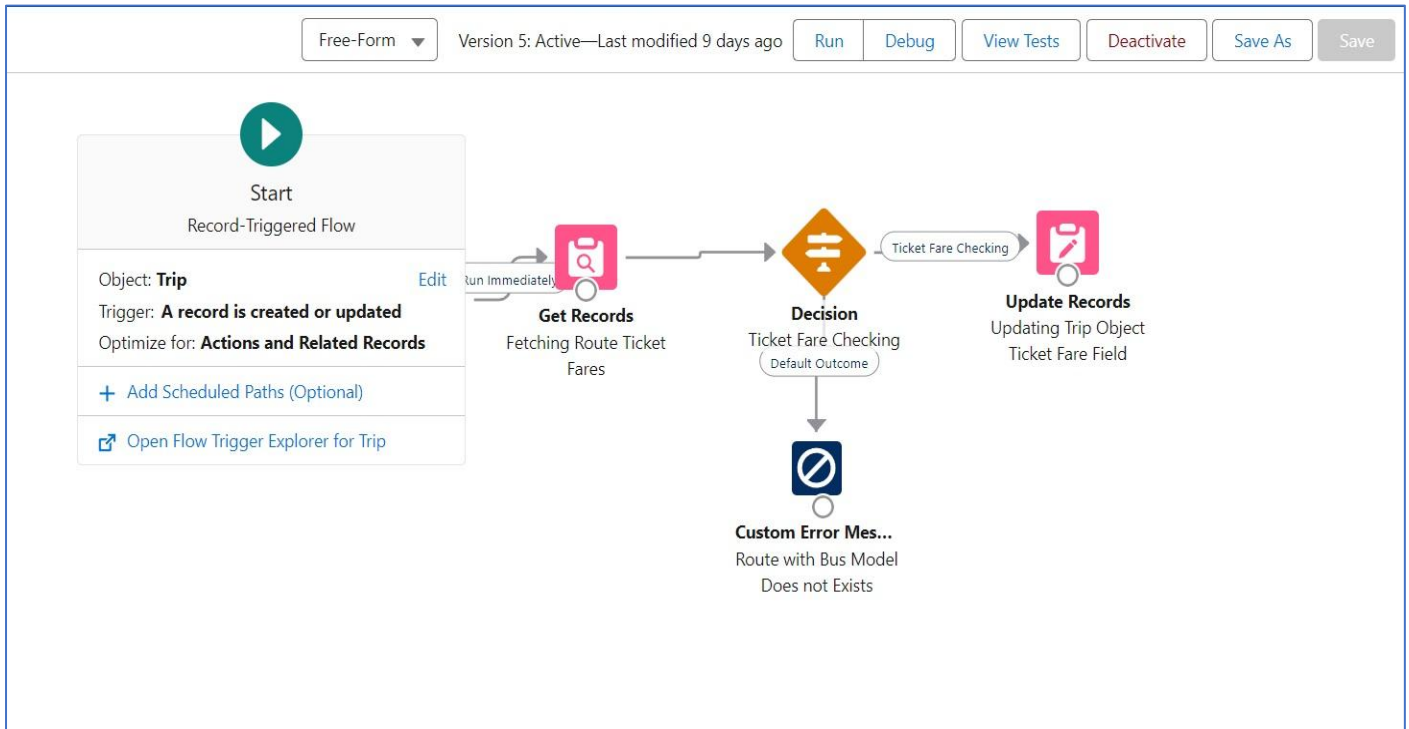
There is no Record with the Route {!\$Record.Route\_Name\_\_r.Name} and the Bus Model {!\$Record.Bus\_No\_\_r.Model\_\_c} in the Ticket Fares

+ Add Error Message

Cancel Done

# A CRM Application For Public Transport Management

26. Click Done
27. Save the flow as "Fetching Ticket Fare For Bus"
28. Activate the flow.



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## Milestone 9 Triggers

Triggers in Salesforce are pieces of Apex code that execute before or after specific database operations, such as insert, update, delete, or undelete. They allow you to perform custom actions on records in Salesforce when certain events occur. Triggers are particularly powerful for enforcing business logic and automating workflows.

**Activity 1 : Create a Trigger to validate whether the Driver Id and Conductor Id are correct or not.**

**Step 1:** Login to salesforce

Log in to your Salesforce account with administrative privileges.

**Step 2:**

i) Navigate to Setup: Once logged in, click on the gear icon ?? (Setup) located at the top-right corner of the page. This will open the Setup menu.

ii) Click on Developer Console: Click on the "Developer Console" option from the Setup menu. This will open the Developer Console in a new browser tab or window.

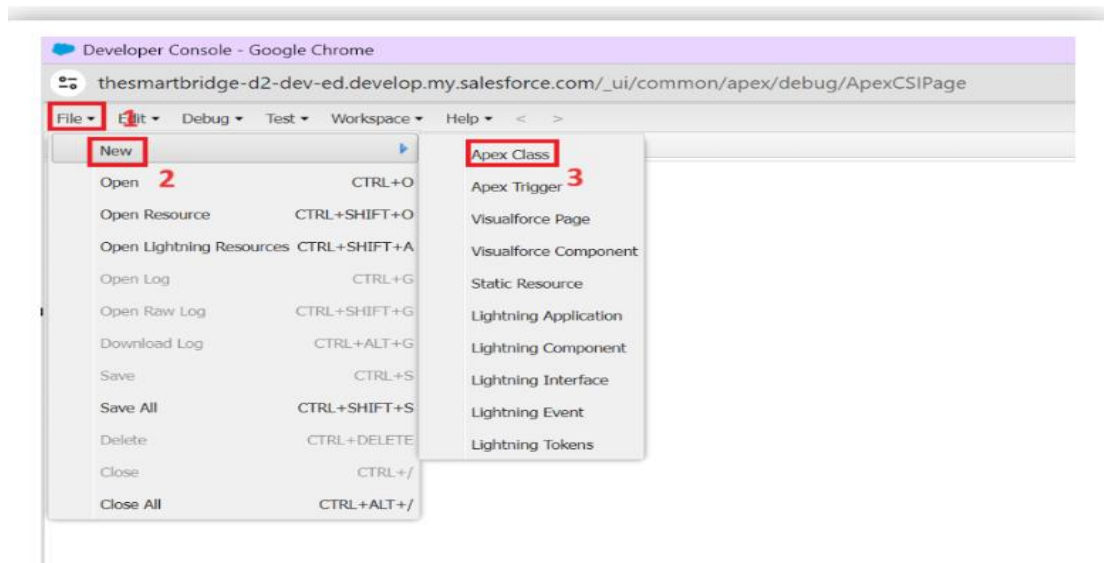
**Step 3:**

1) In the Developer Console window, go to the top menu and click on "File".

2) Select New: From the dropdown menu under "File", select "New".

3) Choose Apex Class: In the submenu that appears, select "Apex Class". This will open a new Apex Class editor tab.

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Give Class Name : TripTriggerHandlerClass

Create an Apex Class:

```
Public class TripTriggerHandlerClass {
```

```
// Checking whether the entered Driver Id belongs to a Driver or not
```

```
public Static void driverValidation(List<Trip__c> tripList){
```

```
    List<Employee__c> driverList = [SELECT Id, Name FROM Employee__c WHERE Role__c ='Driver'  
];
```

```
    If(driverList != null){
```

```
        Map<Id, String> driverMap = new Map<Id, String>();
```

```
        for(Employee__c emp : driverList ){
```

```
            driverMap.put(emp.Id, emp.Name);
```

```
        }
```

```
        for(Trip__c trip : tripList ){
```

```
            If(trip.Driver_Id__c!=null){
```

```
                Boolean hasDriverId = driverMap.containsKey(trip.Driver_Id__c); // hasDriverId will be
```

```
true
```

```
                If(hasDriverId == false){
```

```
                    trip.addError('The assigned person is not a Driver.');
```

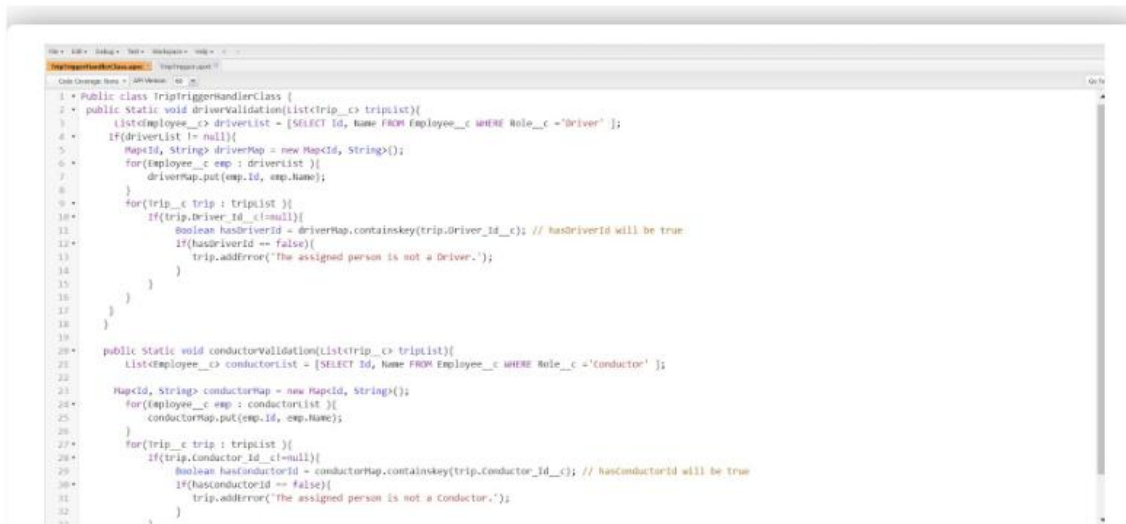
# A CRM Application For Public Transport Management

```
    }
  }
}
}
}

// Checking whether the entered conductor Id belongs to a Conductor or not
public Static void conductorValidation(List<Trip__c> tripList){
    List<Employee__c> conductorList = [SELECT Id, Name FROM Employee__c WHERE Role__c
='Conductor' ];

    Map<Id, String> conductorMap = new Map<Id, String>();
    for(Employee__c emp : conductorList ){
        conductorMap.put(emp.Id, emp.Name);
    }
    for(Trip__c trip : tripList ){
        If(trip.Conductor_Id__c!=null){
            Boolean hasConductorId = conductorMap.containsKey(trip.Conductor_Id__c); //
hasConductorId will be true
            If(hasConductorId == false){
                trip.addError('The assigned person is not a Conductor.');
```

# A CRM Application For Public Transport Management



## Step 4:

- i) In the Developer Console window, go to the top menu and click on "File".
- ii) Select New: From the dropdown menu under "File", select "New".
- iii) Choose Apex Class: In the submenu that appears, select "Apex Trigger". This will open a new Apex Trigger editor tab.

## Create an Apex Trigger:

trigger TripTrigger on Trip\_\_c (before insert, before update) {

if(trigger.isBefore){

if(trigger.isInsert || trigger.isUpdate){

// Validating the Conductor Id in Trip is really a Conductor or not

TripTriggerHandlerClass.driverValidation(trigger.new);

// Validating the Conductor Id in Trip is really a Conductor or not

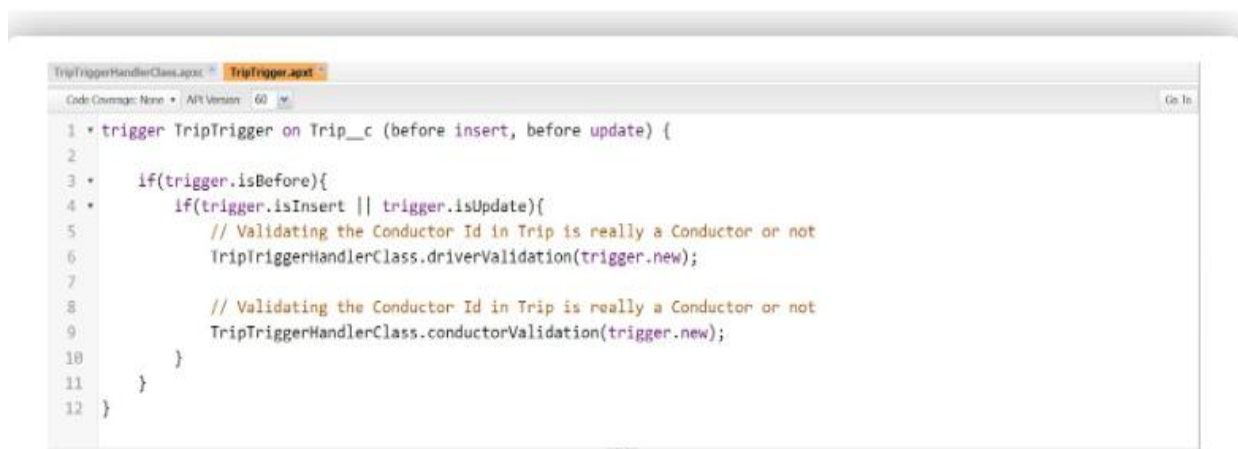
TripTriggerHandlerClass.conductorValidation(trigger.new);

}

}

}

# A CRM Application For Public Transport Management



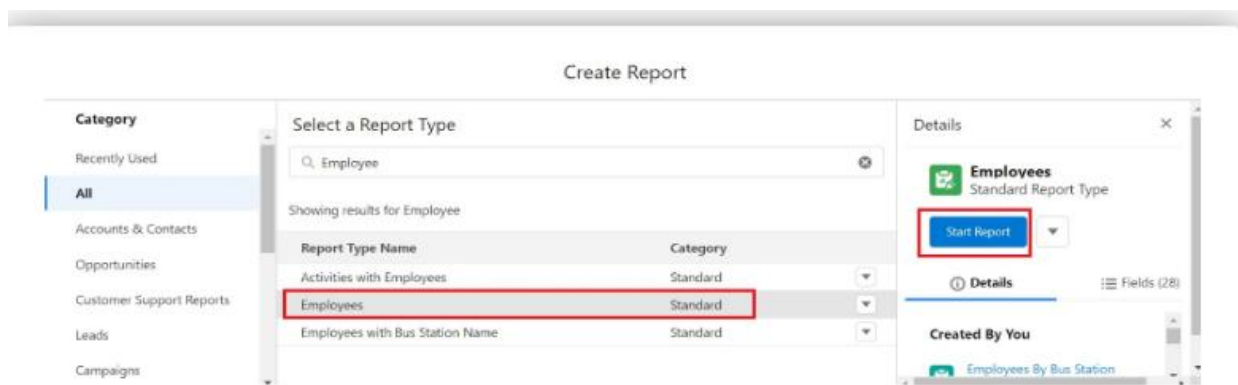
# A CRM Application For Public Transport Management

## Milestone 10 Reports

Reports in Salesforce are tools that allow you to analyze and present your Salesforce data in a structured format. They help you understand and monitor key metrics and trends, providing insights into your business operations. Salesforce reports are highly customizable and can be tailored to meet specific business requirements.

### Activity 1: Create a Employees By Bus Station(Summary) Report

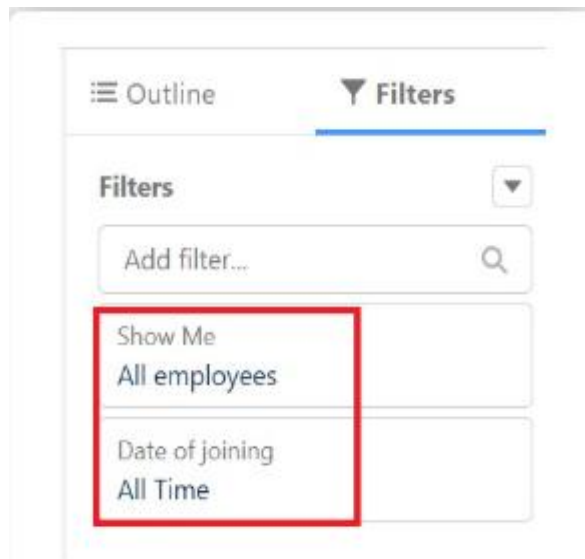
1. Click App Launcher
2. Select Public Transport(RTC) App
3. Click on Reports tab
4. Click on New Report.
5. Click the report type as Employees Click Start report.





# A CRM Application For Public Transport Management

6. Click on Filters and select as follows and click on Apply



7. Customize your report, in group rows select – Bus Station Name, for columns Employee ID, Employee Name, Role (In this way we are making a Summary Report).

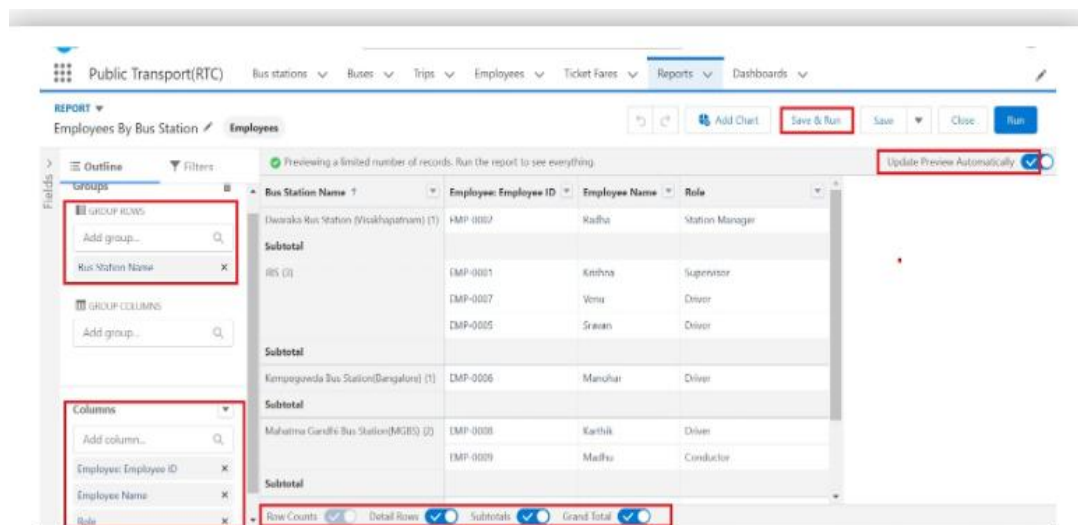
8. Click save and run

9. Give report name – Employees By Bus Station

10. Click Save

NOTE: In this report you can see your all record of the object you selected for reporting (What you selects in "Select a report type option")

# A CRM Application For Public Transport Management



## View Report

1. Click on App Launcher on the left side of the screen.
2. Search Public Transport(RTC) App & click on it.
3. Click on Reports Tab.
4. Click on Employees By Bus Station and see records.

Report: Employees  
**Employees By Bus Station**

Enable Field Editing

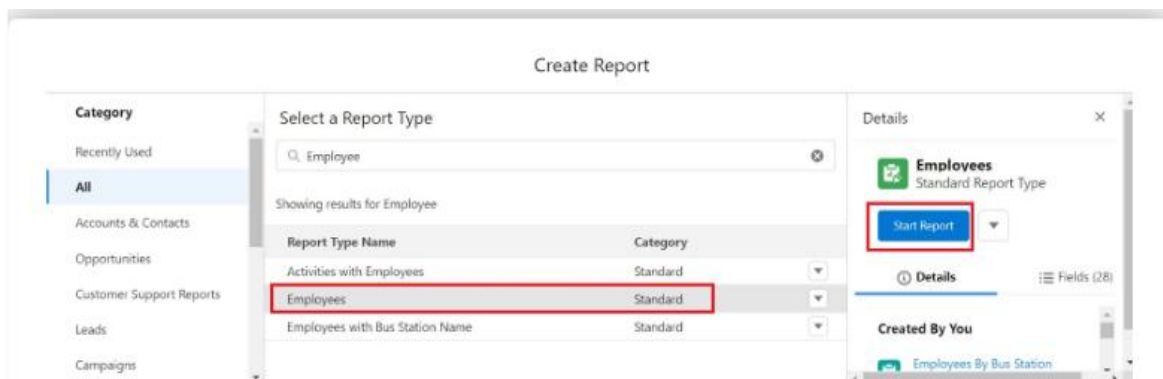
| Bus Station Name ↑                      | Employee: Employee ID | Employee Name | Role                            |
|-----------------------------------------|-----------------------|---------------|---------------------------------|
| Dwaraka Bus Station (Visakhapatnam) (1) | EMP-0002              | Radha         | Station Manager                 |
| Subtotal                                |                       |               |                                 |
| JBS (3)                                 | EMP-0005              | Pravan        | Driver                          |
|                                         | EMP-0007              | Venu          | Driver                          |
|                                         | EMP-0001              | Krishna       | Supervisor                      |
| Subtotal                                |                       |               |                                 |
| Kempegowda Bus Station(Bangalore) (1)   | EMP-0006              | Manohar       | Driver                          |
| Subtotal                                |                       |               |                                 |
| Mahatma Gandhi Bus Station(MGBS) (2)    | EMP-0009              | Madhu         | Conductor                       |
|                                         | EMP-0008              | Karthik       | Driver                          |
| Subtotal                                |                       |               |                                 |
| Srinivasa Bus Station (Tirupathi) (2)   | EMP-0003              | Karthik       | Customer Service Representative |

Row Counts ☒ Detail Rows ☒ Subtotals ☒ Grand Total ☒

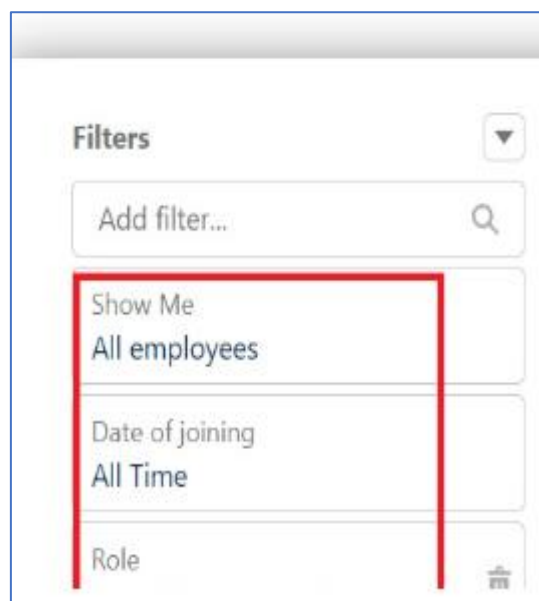
# A CRM Application For Public Transport Management

## Activity 2: Create a Drivers and Conductors Information Report

1. Click App Launcher
2. Select Public Transport(RTC) App
3. Click on Reports tab
4. Click on New Report.
5. Click the report type as Employees Click Start report.



6. Click on Filters and select as follows and click on Apply



# A CRM Application For Public Transport Management

7. Customize your report, in group rows select – Bus Station Name, for columns Employee ID, Employee Name, Role (In this way we are making a Summary Report).
8. Click save and run
9. Give report name – Drivers And Conductors Information
10. Click Save

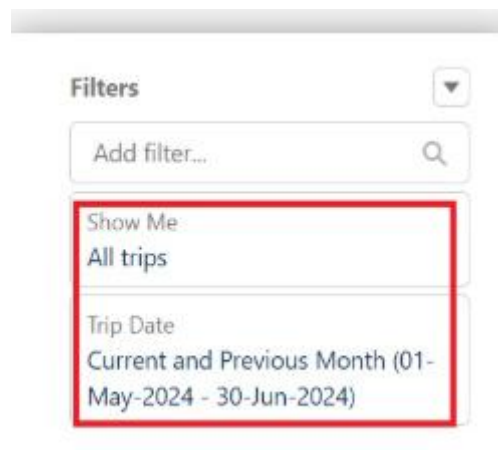
The screenshot shows the 'Public Transport(RTC)' application interface. The top navigation bar includes 'Bus stations', 'Buses', 'Trips', 'Employees', 'Ticket Fares', 'Reports', and 'Dashboards'. The 'Reports' tab is active, showing a report titled 'Drivers and Conductors Information' based on the 'Employees' data source. The report is configured with 'Bus Station Name' as the group row and 'Employee ID', 'Employee Name', and 'Role' as columns. The report is saved and ready to be run. The report preview shows a table with columns: Bus Station Name, Employee ID, Employee Name, and Role. The data is grouped by Bus Station Name, showing details for JBS (2), Kempegowda Bus Station(Bangalore) (1), Mahatma Gandhi Bus Station(MGBS) (2), and Srinivasa Bus Station (Tirupathi) (1). The report is saved and ready to be run.

| Bus Station Name                      | Employee ID | Employee Name | Role      |
|---------------------------------------|-------------|---------------|-----------|
| JBS (2)                               | EMP-0007    | Venu          | Driver    |
|                                       | EMP-0005    | Sravan        | Driver    |
| Subtotal                              |             |               |           |
| Kempegowda Bus Station(Bangalore) (1) | EMP-0006    | Manohar       | Driver    |
| Subtotal                              |             |               |           |
| Mahatma Gandhi Bus Station(MGBS) (2)  | EMP-0008    | Karthik       | Driver    |
|                                       | EMP-0009    | Madhu         | Conductor |
| Subtotal                              |             |               |           |
| Srinivasa Bus Station (Tirupathi) (1) | EMP-0004    | Bhuvan        | Driver    |
| Subtotal                              |             |               |           |
| Total (6)                             |             |               |           |

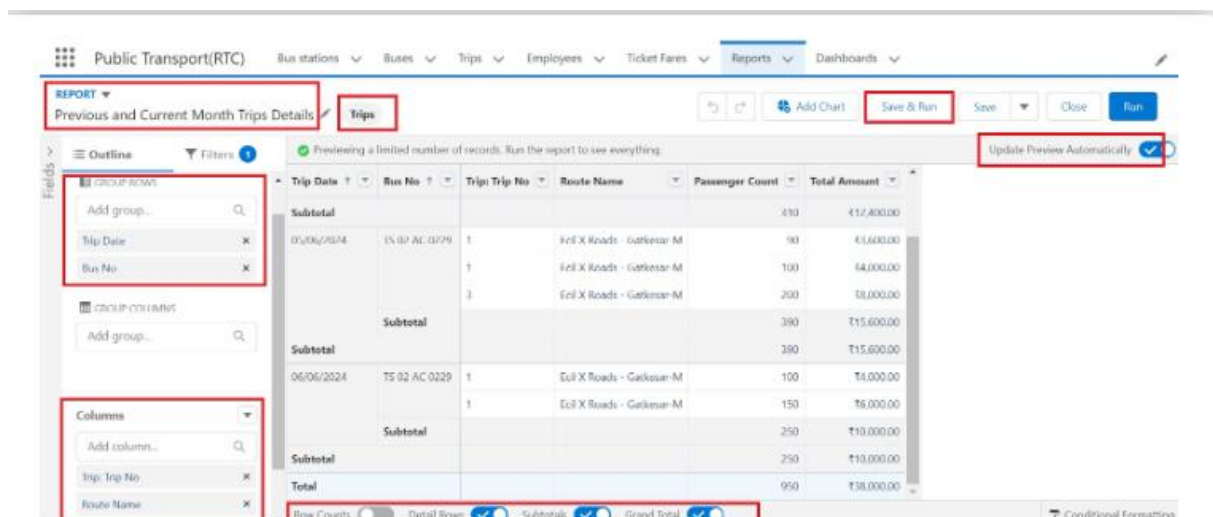
## Activity 3: Create a Previous and Current Month Trip Details Report

1. Click App Launcher
2. Select Public Transport(RTC) App
3. Click on Reports tab
4. Click on New Report.
5. Click the report type as Trips Click Start report.
6. Click on Filters and select as follows and click on Apply

# A CRM Application For Public Transport Management



7. Customize your report, in group rows select – Trip Date, Bus No , for columns Trip No, Route Name, Passenger Count, Total Amount (In this way we are making a Summary Report).
8. Click save and run
9. Give report name – Previous And Current Month Trips Details
10. Click Save



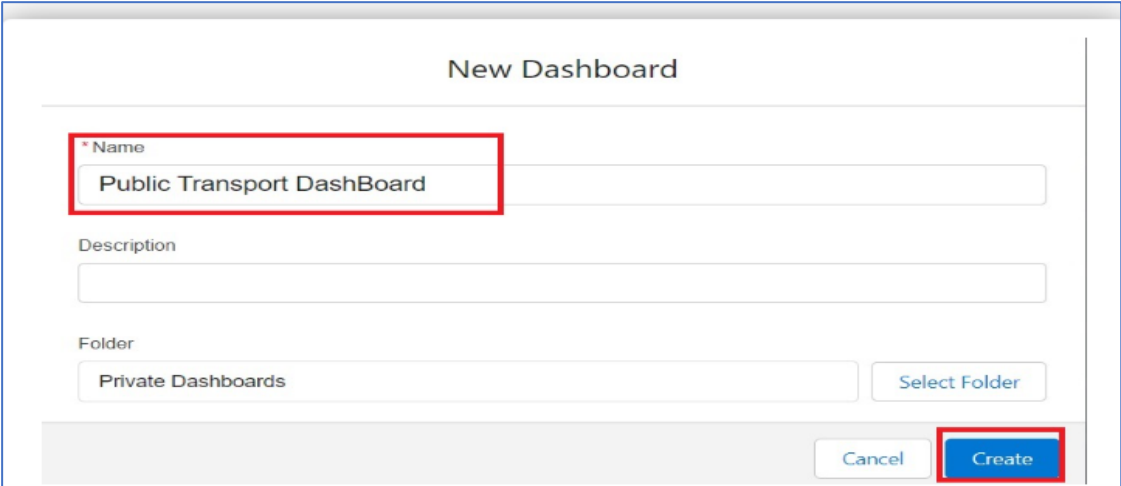
# A CRM Application For Public Transport Management

## Milestone 11 Dashboards

Dashboards in Salesforce are visual representations of your reports and key metrics, providing a consolidated view of your data. They allow you to monitor performance, track progress, and make informed decisions at a glance. Dashboards are composed of various components such as charts, tables, gauges, and metrics, each displaying data from one or more reports.

### Activity 1: - Create Dashboard

1. Click on the Dashboards tab from the Public Transport(RTC) application.
2. Click on the new dashboard.

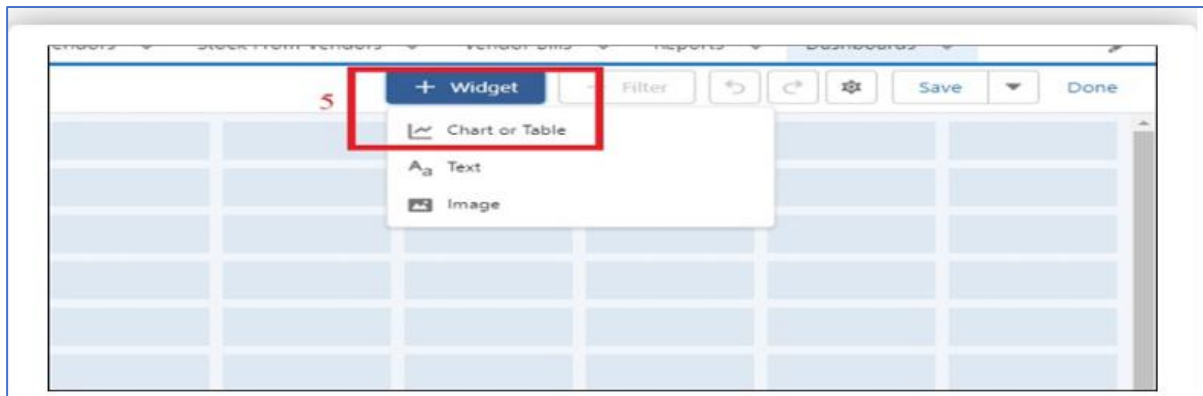


The screenshot shows the 'New Dashboard' form in Salesforce. The form is titled 'New Dashboard' and contains the following fields and buttons:

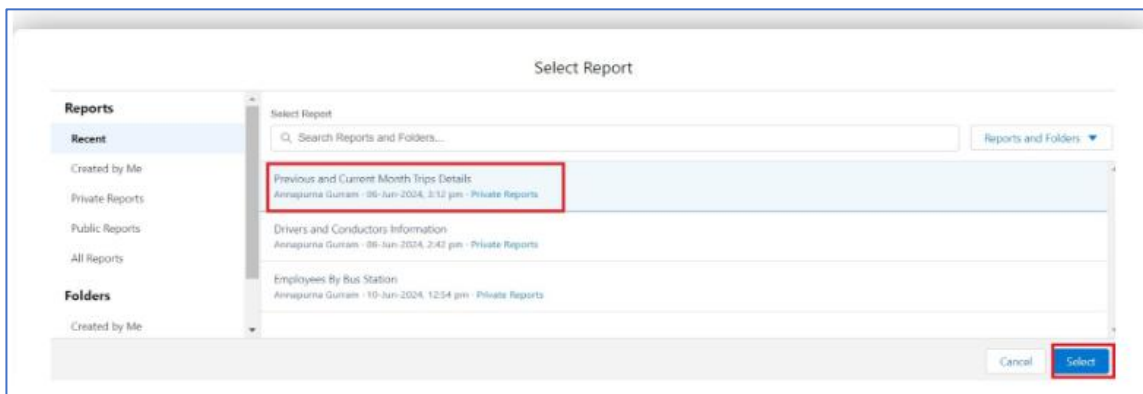
- Name:** A text field containing 'Public Transport DashBoard', highlighted with a red box.
- Description:** An empty text field.
- Folder:** A dropdown menu showing 'Private Dashboards' and a 'Select Folder' button.
- Buttons:** 'Cancel' and 'Create' buttons at the bottom right, with the 'Create' button highlighted by a red box.

3. Give name - Public Transport DashBoard
4. Click create
5. Click on +widget
6. Select the Previous and Current Month Trips Details Report
7. For the data visualization select any of the charts, tables etc. as per your choice/requirement
8. Click add.
9. Click on +widget

# A CRM Application For Public Transport Management



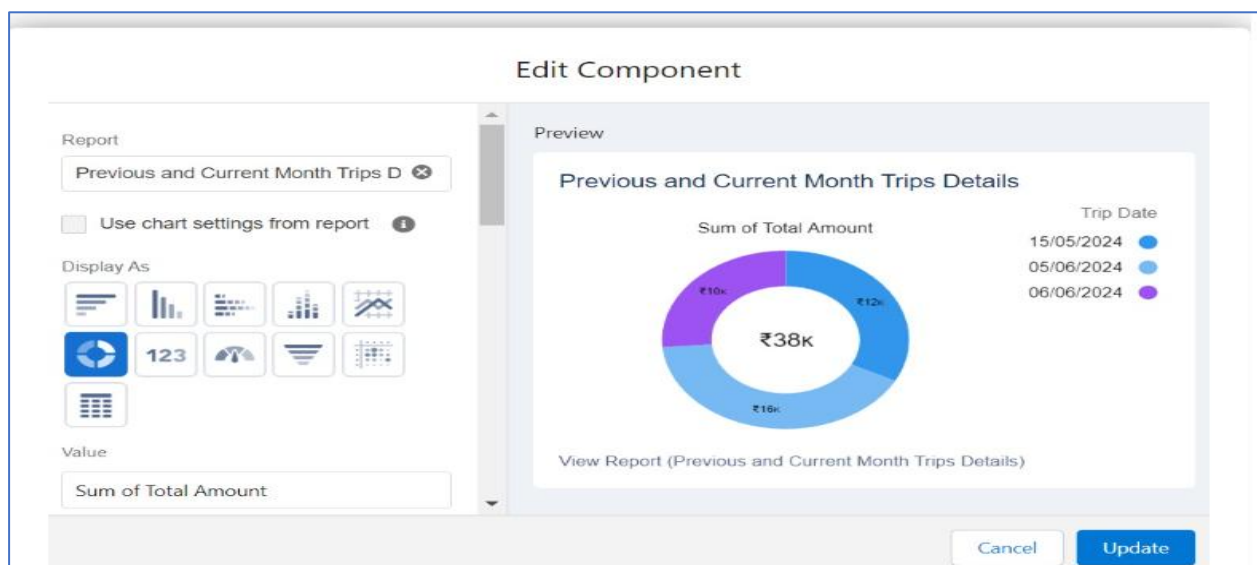
10. Select the Drivers and Conductors Information Report



11. For the data visualization select any of the charts, tables etc. as per your choice/requirement

12. Click add.

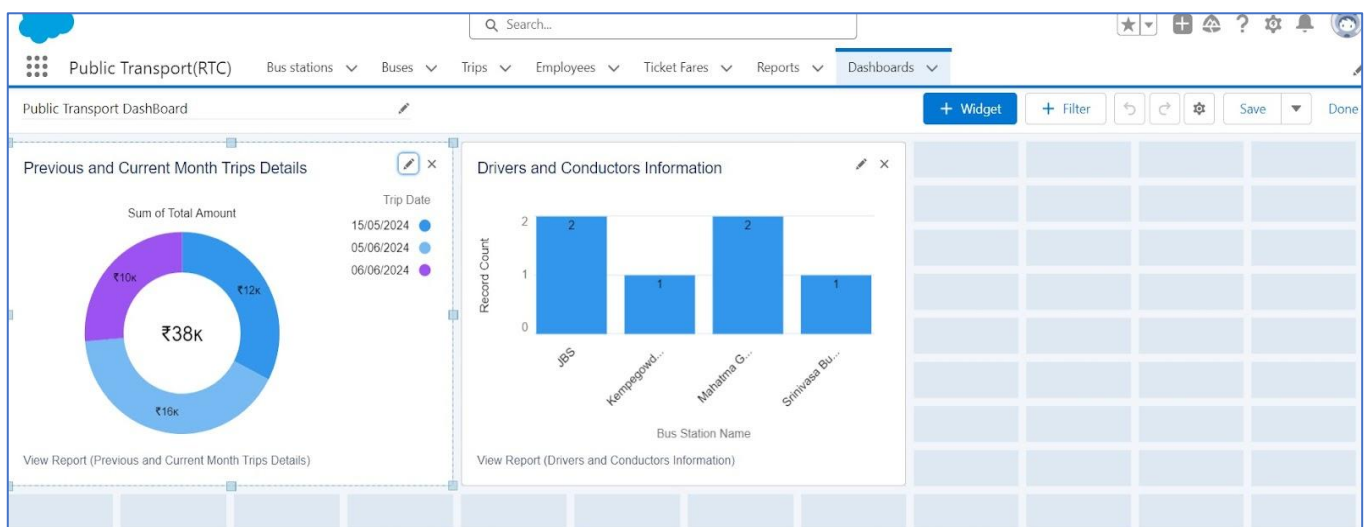
13. Click save.



# A CRM Application For Public Transport Management

## Activity 2: View Dashboard

1. Click on App Launcher on the left side of the screen.
2. Search Public Transport(RTC) & click on it.
3. Click on Dashboard Tab.
4. Click on Public Transport Dashboard see graph view of records





# A CRM Application For Public Transport Management

## OUT PUTS:

### Bus station:-

New Bus Station

\* = Required Information

Information

\* Bus Station Name

Shelter available

☐

Amenities

Please fill out this field.

Available

Accessibility

Waiting Area

Information and S...

Food and Drink

Chosen

Bench

☐

Bus Stop Category

--None--

Owner

 sanjay polisetty

- Address Information:

Address Information

Street

State/Province

\* City

Zip/PostalCode

Location

Cancel

Save & New

Save

Page | 73

# A CRM Application For Public Transport Management

## Buses:

New Bus

\* = Required Information

Information

Bus Station Name

Search Bus Stations...

Owner

sanjay polisetty

\* Category

--None--

View all dependencies

Model

--None--

View all dependencies

\* Capacity

\* Bus Registration No

Cancel

Save & New

Save

## Employees:

- Information:

New Employee

\* = Required Information

Information

\* Employee Id

Work Place

Bus Station Name

Search Bus Stations...

\* Role

--None--

Salary

Date of Joining

Owner

sanjay polisetty

# A CRM Application For Public Transport Management

- Personal Details:

Personal Details

Date of Birth

Phone

- Address:

Address

Street

City

State/Province

Country

Zip/PostalCode

Cancel

Save & New

Save

# A CRM Application For Public Transport Management

## Ticket fares:

New Ticket Fare

\* = Required Information

Information

\* Route Name

Owner

 sanjay polisetty

Ticket Fare

Bus Model

--None--

Cancel

Save & New

Save

- Information:

\* Trip Date



Format: 12/31/2024

\* Trip No

\* Bus No

Search Buses...



Driver Id

Search Employees...



Conductor Id

Search Employees...



Owner

 sanjay polisetty

- Bus Schedule:

Bus Schedule

\* Route Name

Search Ticket Fares...



\* Bus Starting Terminal

Departure Time

--None--

No. of Stops

Estimated Travel Time

\* Destination Terminal

Arrival Time

--None--

Frequency Per Day

# A CRM Application For Public Transport Management

- Passenger Information:

Passenger Information

\* Passenger Count

Ticket Fare

Cancel

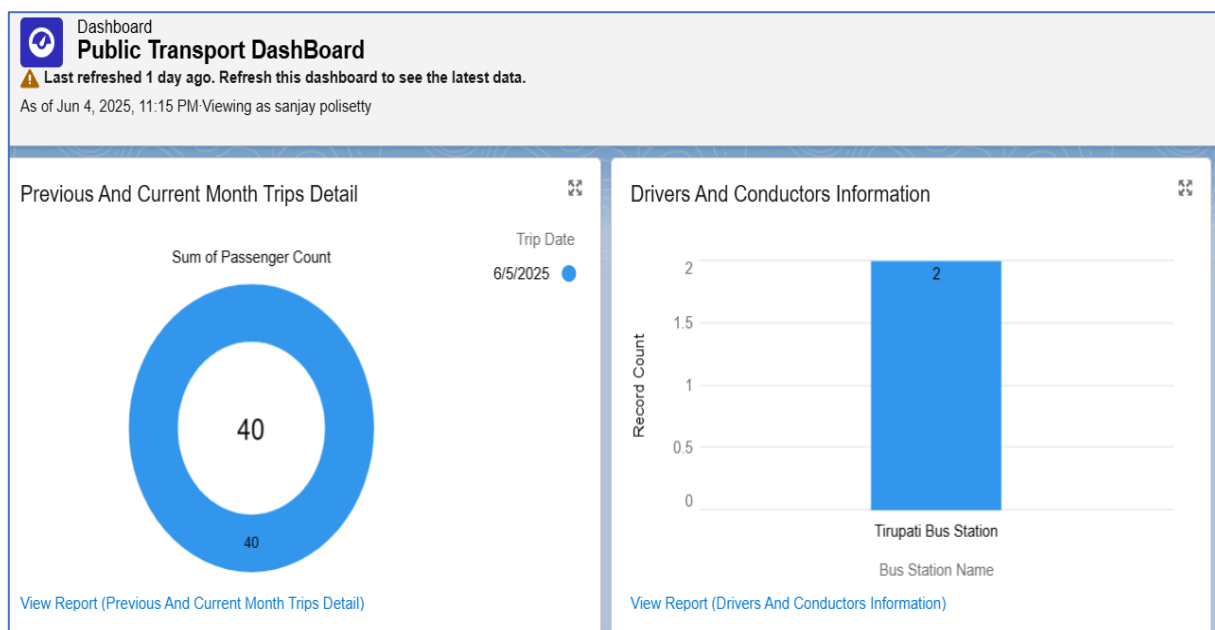
Save & New

Save

## Reports:

| Reports         |                                         |                                                                                                           |                 |                            |                     |            |            |
|-----------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------|----------------------------|---------------------|------------|------------|
| Recent          |                                         | 4 items                                                                                                   |                 | Q Search recent reports... |                     | New Report | New Folder |
| REPORTS         | Report Name                             | Description                                                                                               | Folder          | Created By                 | Created On          | Subscribed |            |
| Recent          | Previous And Current Month Trips Detail |                                                                                                           | Private Reports | sanjay polisetty           | 6/4/2025, 9:13 PM   |            |            |
| Created by Me   | Employees By Bus Station                |                                                                                                           | Private Reports | sanjay polisetty           | 6/3/2025, 11:42 PM  |            |            |
| Private Reports | Drivers And Conductors Information      |                                                                                                           | Private Reports | sanjay polisetty           | 6/4/2025, 9:06 PM   |            |            |
| Public Reports  | Sample Flow Report: Screen Flows        | Which flows run, what's the status of each interview, and how long do users take to complete the screens? | Public Reports  | Automated Process          | 4/30/2025, 12:24 AM |            |            |
| All Reports     |                                         |                                                                                                           |                 |                            |                     |            |            |
| FOLDERS         |                                         |                                                                                                           |                 |                            |                     |            |            |

## Dashboards:



# A CRM Application For Public Transport Management

## **Milestone 12**

### **Conclusion**

By implementing this Salesforce-based Public Transport( RTC) Management System, the RTC department can significantly improve its operational efficiency, data management, and overall service quality to passengers. Data-driven decision-making capabilities for management. Accurate and up-to-date records of all operational data.

# A CRM Application For Public Transport Management

## **GITHUB LINK:-**

<https://github.com/sanjaypolisettySanju/a-crm-application-for-public-transport-LTVIP2025TMID29546>

## **DEMO VIDEO LINK:-**

[https://drive.google.com/file/d/1vTcubpBCy7HDw4URDYpNjCD2HV9DnoaA/view?usp=drive link](https://drive.google.com/file/d/1vTcubpBCy7HDw4URDYpNjCD2HV9DnoaA/view?usp=drive_link)